MTNAI

Mountain Terrain Analysis — User Guide

Features, how to use them, every setting and how each tool thinks

Built on the teachings of GSP Mountain Warfare (low and high altitude operations)

How to read this guide. Each tool has four parts: what it does (in plain words), the doctrine behind it, a flowchart of how it works, and a table of every setting with its default. Screenshots use the demo terrain.

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1. WHAT MTNAI IS

MtnAI is a map program for planning fights in the mountains. You give it a height map of the ground (a DEM). It then tells you, square by square:

- where troops and vehicles **can go, go slowly, or cannot go**;
- which **heights matter** and where the **passes** are;
- what can be **seen** from a point, and what the **enemy can see**;
- the best places for **posts, OPs, weapons, minefields, picquets and helipads**;
- the best **route** from A to B, how long it takes, and a **route card**.

It runs **fully offline** in a web browser. Nothing leaves the computer. The only optional internet use is the Download DEM tool.

It looks and works like **ArcGIS Pro**: a ribbon at the top, layers on the left, tools on the right, map in the middle.

Doctrine. The rules in GSP Mountain Warfare are built into the tools: march times, ridge routes, tiered defence, strong points, OPs at many heights, minefields in valleys, uphill / downhill fire, picquets and helipad slopes (Section 13 lists them all).

Important. MtnAI suggests; it does not decide. Always confirm on the ground, as the pamphlet directs.

2. QUICK START (FIVE MINUTES)

1. Double-click **MtnAI.html** (or open index.html). Use Chrome, Edge or Firefox.
2. Click **Demo terrain** to load the practice area, or **Add Data** to open your own DEM, or **Download DEM** to fetch one.
3. On the **Map** tab, use **Situation** to place Own position, Enemy, Target, Start A and End B (the demo has them already).
4. On the **Analysis** tab, click a tool (for example **Posts / OPs**). Its settings open on the right. Click **Run**.
5. Look at the result on the map, in the **Attributes** pane and in the **Attribute table**. Save with **Ctrl+S**; print the report with **Ctrl+P**.

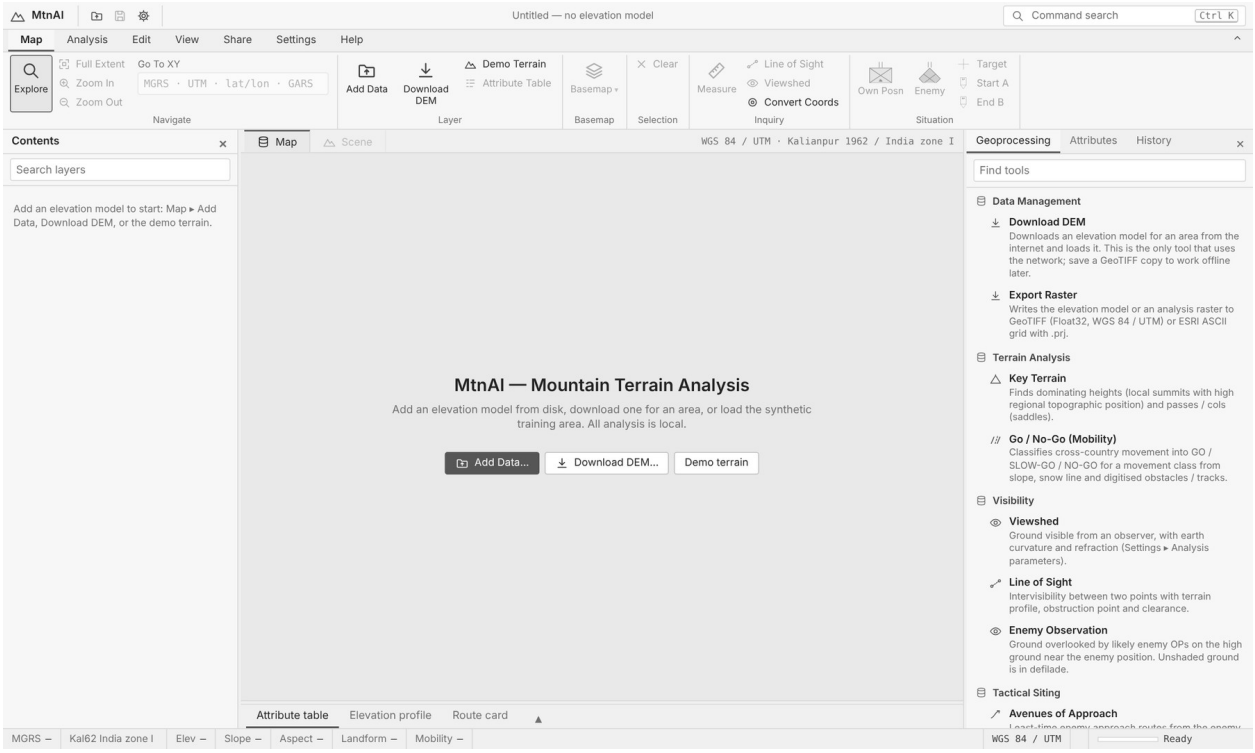


Figure 1. The start screen: Add Data, Download DEM or Demo terrain.

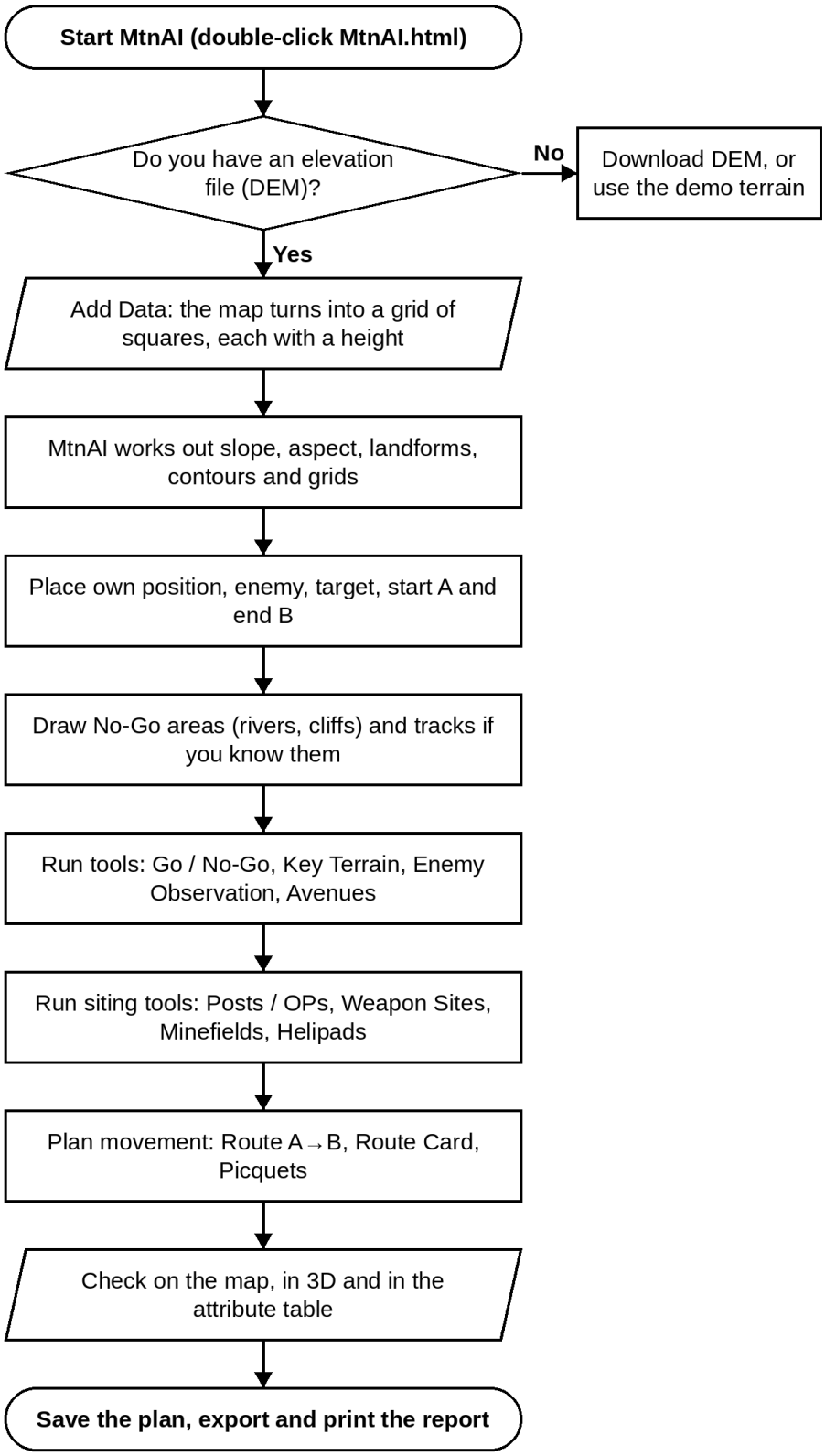


Figure 2. Flowchart — the whole planning job, start to finish.

3. THE SCREEN

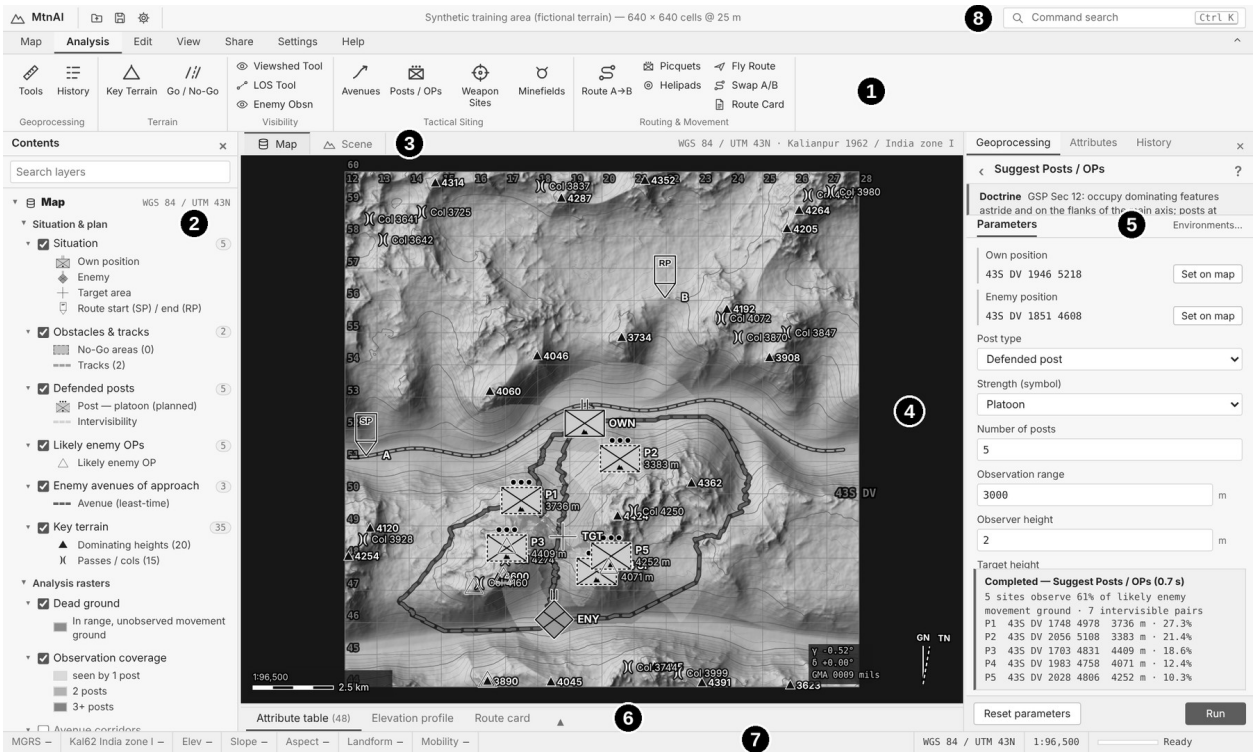


Figure 3. The MtnAI screen. The numbers match the table below.

No.	Part	What it is for
1	Ribbon	Tabs of buttons: Map, Analysis, Edit, View, Share, Settings, Help. Double-click a tab to hide or show it.
2	Contents pane	Every layer with a tick box (show / hide) and a key. Right-click a layer for Zoom To, Attribute Table, Transparency, Remove, Properties.
3	View tabs	Map (flat, 2D) or Scene (3D). The text on the right shows the coordinate systems in use.
4	Map	Drag to move, wheel to zoom, click a symbol to see its details.
5	Side pane	Geoprocessing (tool settings and Run), Attributes (details of what you clicked), History (every tool run).
6	Bottom dock	Attribute table, Elevation profile and Route card.
7	Status bar	Grid reference, height, slope, aspect, landform and Go / No-Go under the mouse; scale; progress.
8	Command search	Type what you want (Ctrl+K), for example "minefield".

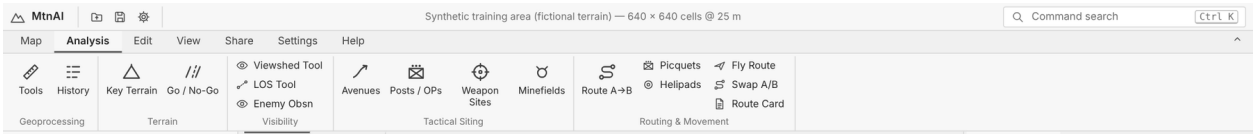


Figure 4. The Analysis tab of the ribbon: every planning tool is one click away.

Running any tool

1. Click the tool on the ribbon (or search for it). Its form opens in the **Geoprocessing** pane.
2. Read the **Doctrine** note at the top (click it to open it fully).
3. Check the settings. Points can be **picked on the map** or typed as a grid reference. Click a grey section heading to open more settings.
4. Click **Run**. A green note shows the result; red means it failed and says why.
5. **Reset parameters** puts every setting back to its default. Your changes are remembered between sessions.

4. GETTING TERRAIN (THE DEM)

A DEM is a picture where each pixel is a height. MtnAI turns it into a grid of equal squares (cells) in metres (WGS 84 / UTM).

Add Data (offline)

- Files: SRTM **.hgt**, **GeoTIFF** (.tif), ESRI **ASCII grid** (.asc) or a **.zip** of these. Several tiles join together.
- Drag files onto the map, or Map ▸ **Add Data** (Ctrl+O).
- **Crop to AOI** (Edit tab) cuts out a smaller area at finer detail. **Full DEM** goes back.

Download DEM (needs internet)

Fetches heights for any area. Save the GeoTIFF copy it makes, then you can work offline later with Add Data.

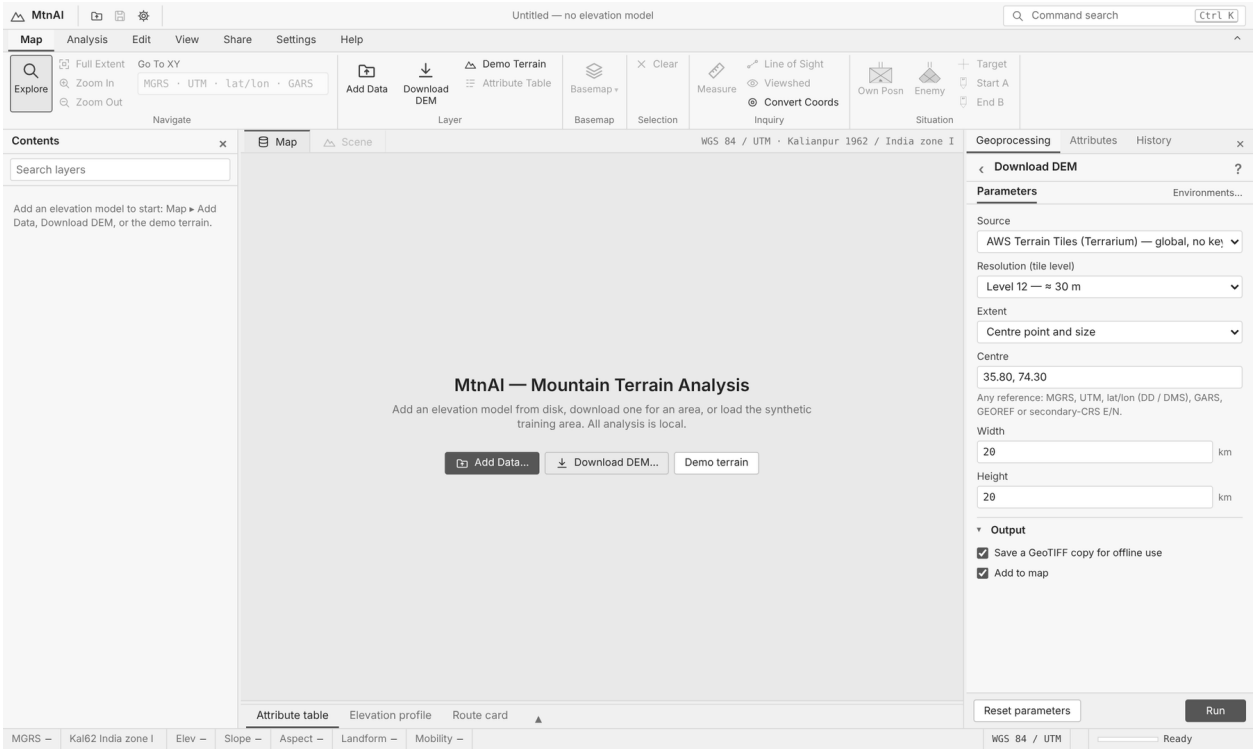


Figure 5. The Download DEM tool.

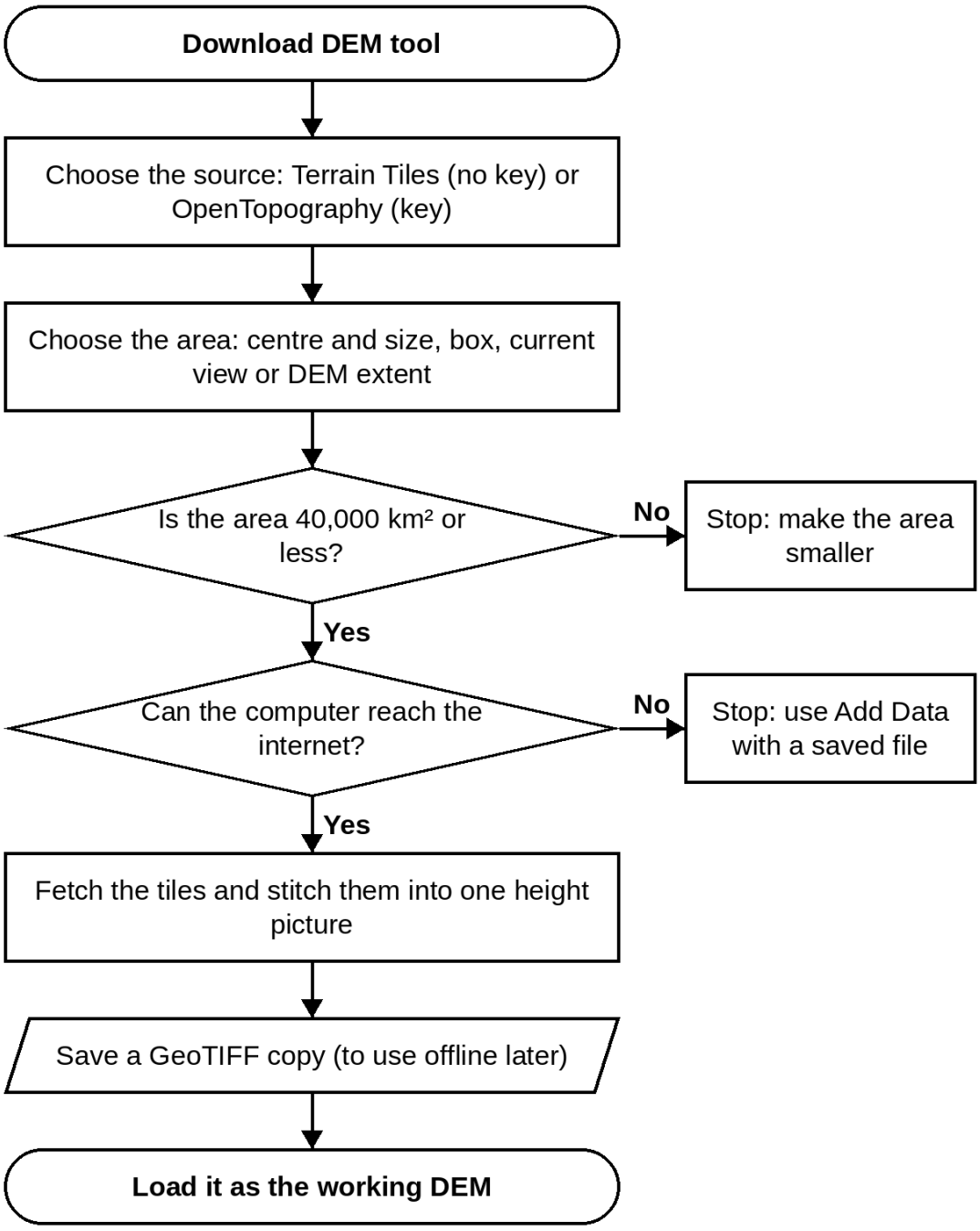


Figure 6. Flowchart — Download DEM.

Setting	Default	Notes
Source	AWS Terrain Tiles (Terrarium) — global, no key	
Resolution (tile level)	Level 12 — ≈ 30 m	only shown for some choices
Dataset	Copernicus GLO-30 (30 m)	only shown for some choices

Setting	Default	Notes
OpenTopography API key	—	Free from opentopography.org (MyOpenTopo ▸ Get an API key). only shown for some choices
Extent	Centre point and size	
Centre	35.80, 74.30	Any reference: MGRS, UTM, lat/lon (DD / DMS), GARS, GEOREF or secondary-CRS E/N. only shown for some choices
Width	20 km	only shown for some choices
Height	20 km	only shown for some choices
West (lon °)	74.2	only shown for some choices
South (lat °)	35.7	only shown for some choices
East (lon °)	74.45	only shown for some choices
North (lat °)	35.9	only shown for some choices
Output		
Save a GeoTIFF copy for offline use	Yes	
Add to map	Yes	

Export Raster

Writes the height map, slope, aspect, landforms, ruggedness or any analysis layer to a GeoTIFF or ASCII grid, for use in other GIS programs.

Setting	Default	Notes
Raster	Elevation	
Format	GeoTIFF (.tif)	

5. COORDINATES, GRIDS AND MEASURING

Open **Settings ▸ Coordinate Systems** (or click the coordinate text above the map).

- **Analysis grid:** WGS 84 / UTM, zone chosen automatically or forced.
- **Second system:** for example Kalianpur 1962 / India zone I or IIa, Kalianpur UTM, or your own.
- **Grid references:** MGRS (2 to 10 figures), UTM, latitude / longitude (DD, DM, DMS), GARS, GEOREF.
- **North:** grid, true or magnetic (enter declination). Angles in mils (6400 or 6000), degrees or gradians.
- **Map grids:** MGRS / UTM lines, a lat / long graticule and the second system's grid (View ▸ Grids & Labels).

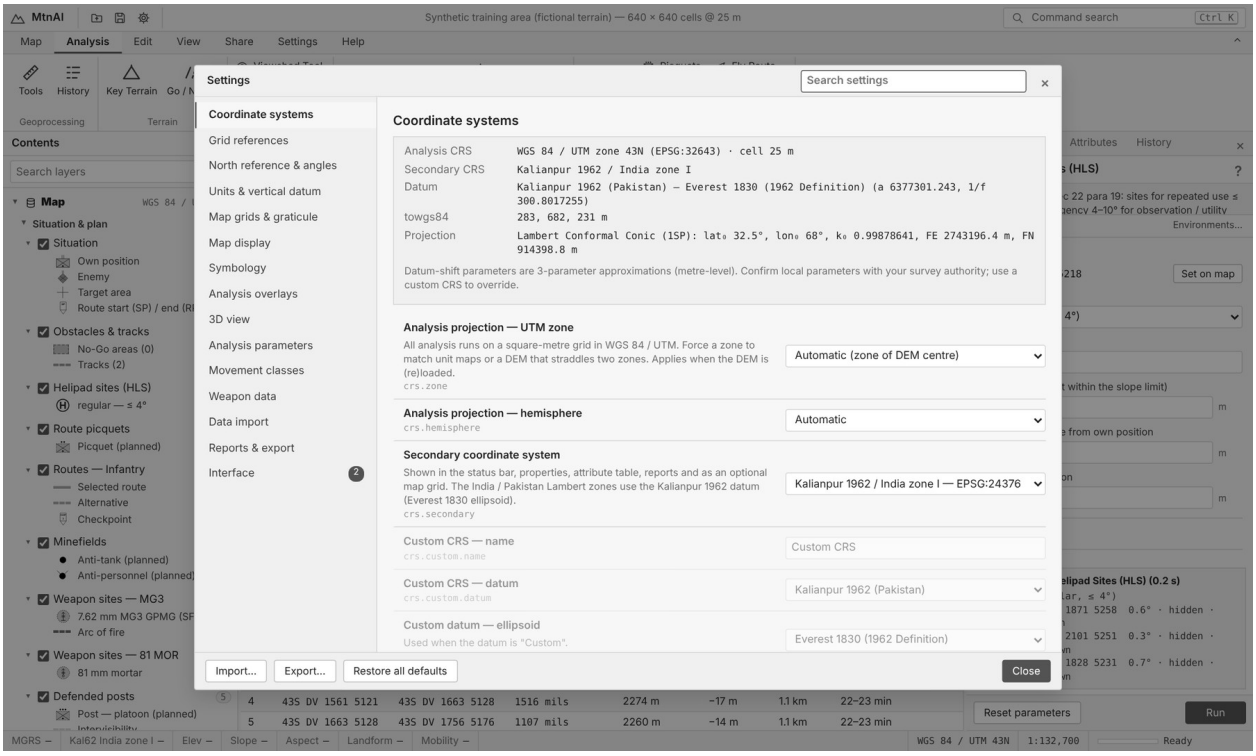


Figure 7. Settings ▸ Coordinate systems.

Go To XY and Convert Coordinates

Type any reference in **Go To XY** (Map tab) and press Enter to jump there. **Convert Coords** shows the same point in every format.

Measure

Map ▸ **Measure** (key M): click two points. You get distance, slope distance, grid / true / magnetic bearing, back bearing, height difference and the **GSP march time** by day and by night.

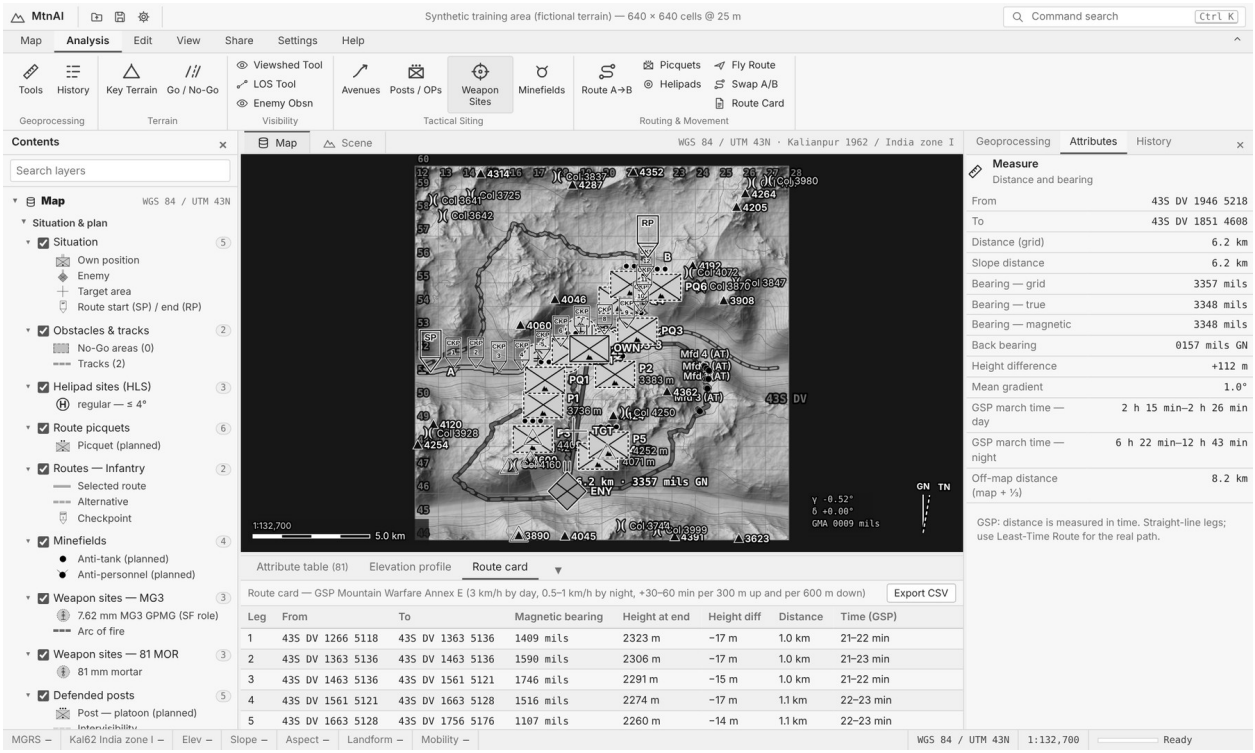


Figure 8. Measure result with the GSP march time.

6. SETTING THE SCENE

Place the units

Button	Meaning	Used by
Own Posn	Your defended area or HQ	Posts, weapons, minefields, avenues, helipads, access
Enemy	Where the enemy is	Enemy observation, avenues, posts, minefields, routes
Target	Area for mortars and guns to hit	Indirect-fire weapon sites
Start A / End B	Start and end of a move	Route A→B, route card, picquets

Click the button, then click the map. **Edit ▸ Situation** also sets your echelon and the enemy type (infantry, wheeled, armour, mule).

Draw what the DEM does not show

- **No-Go Area:** click corners of a river, glacier, cliff or minefield; double-click (or Enter) to finish. Backspace removes the last corner, Esc cancels.
- **Track:** click along a road or mule track; double-click to finish. Tracks are always GO and faster.

You see the shape while you draw it. Every tool respects these shapes.

7. TERRAIN TOOLS

7.1 Go / No-Go (Mobility)

Colours every square: **GO** (easy), **SLOW-GO** (hard), **NO-GO** (impossible) for the chosen movement class. It works from slope, snow line, your No-Go areas and tracks.

Doctrine. GSP Sec 4 para 8–13: mobility is the key-note; vehicles keep to valley roads, cross-compartment travel is often impossible. A loaded mule can climb any slope a man can negotiate without using his hands (Annex H).

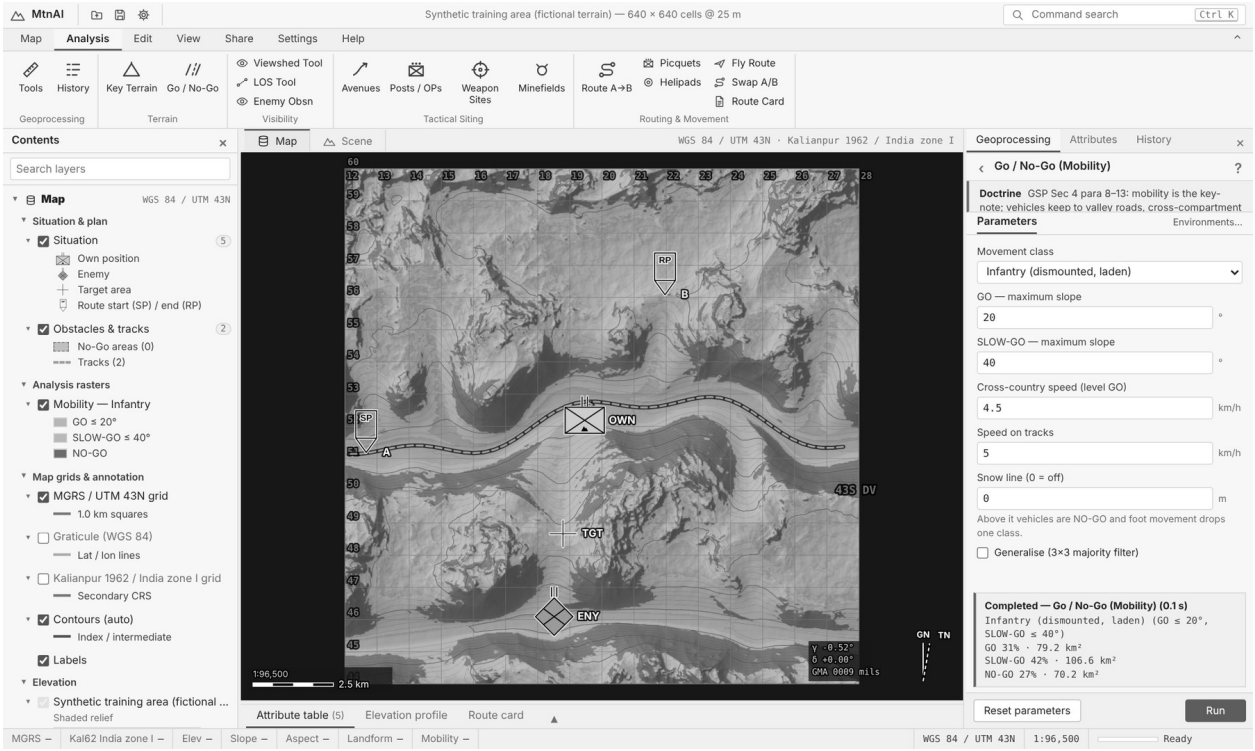


Figure 9. Go / No-Go for infantry. The key for the three classes is in the Contents pane.

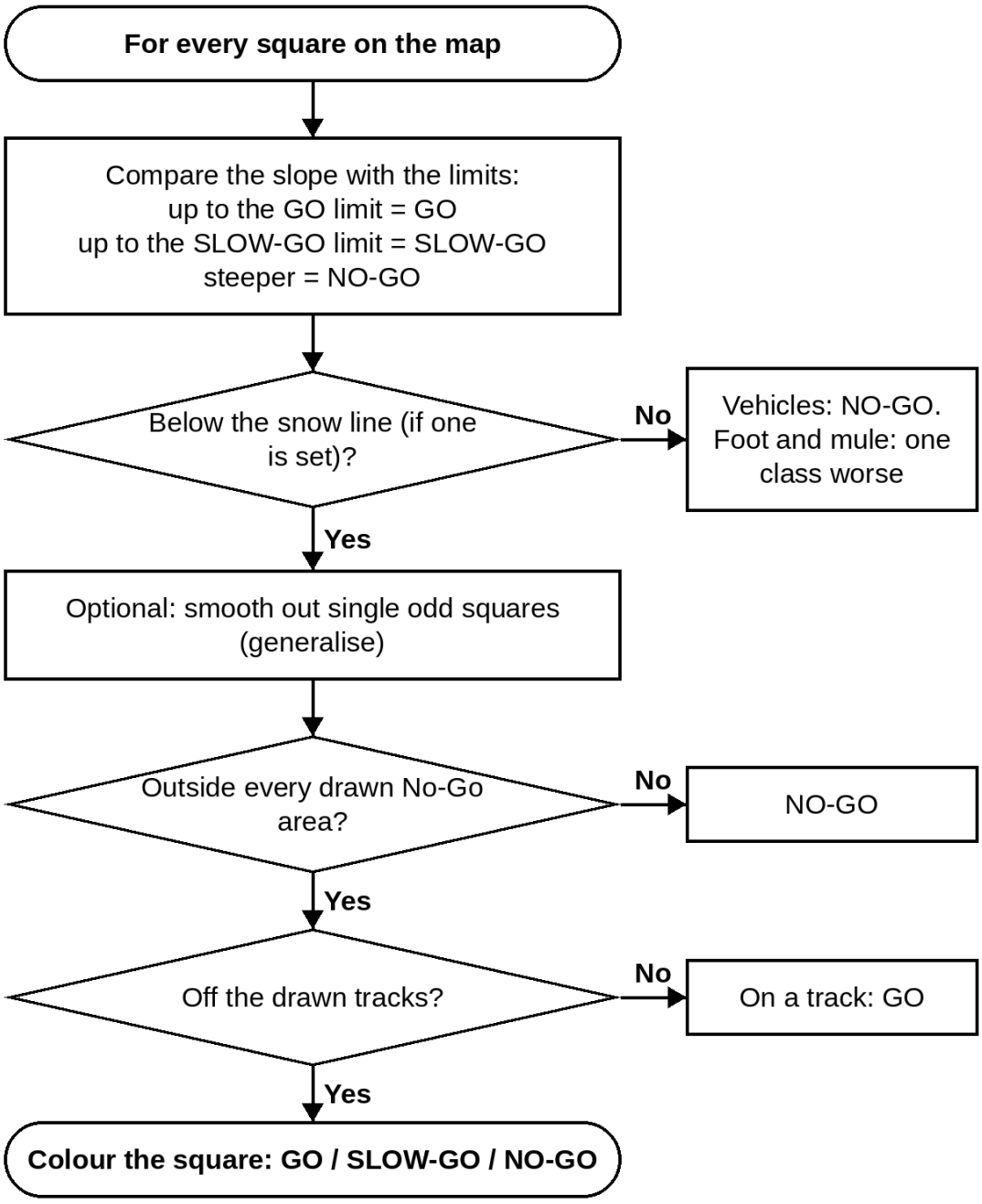


Figure 10. Flowchart — Go / No-Go.

Setting	Default	Notes
Movement class	Infantry (dismounted, laden)	
GO — maximum slope	from movement class	
SLOW-GO — maximum slope	from movement class	
Cross-country speed (level GO)	from movement class	

Setting	Default	Notes
Speed on tracks	from movement class	
Snow line (0 = off)	0 m	Above it vehicles are NO-GO and foot movement drops one class.
Generalise (3×3 majority filter)	No	

Movement class	GO up to	SLOW-GO up to	Speed (km/h)	On tracks (km/h)
Infantry (dismounted, laden)	20°	40°	4.5	5
Mule / Animal Transport	15°	30°	3.5	4.5
Light vehicle (jeep / 4x4)	10°	20°	20	35
Truck / towed gun	6°	12°	12	30
Tracked (tank / APC)	15°	30°	18	30

7.2 Key Terrain

Finds **dominating heights** (summits that stand high above the area around them) and **passes / cols** (low points between two heights that lead from one valley to the next).

Doctrine. GSP Sec 4 para 6–7: routes are overlooked by dominating heights; at brigade level and below, objectives are the seizure or retention of specific dominating heights. Passes are held from the heights on both shoulders (Sec 12 para 14).

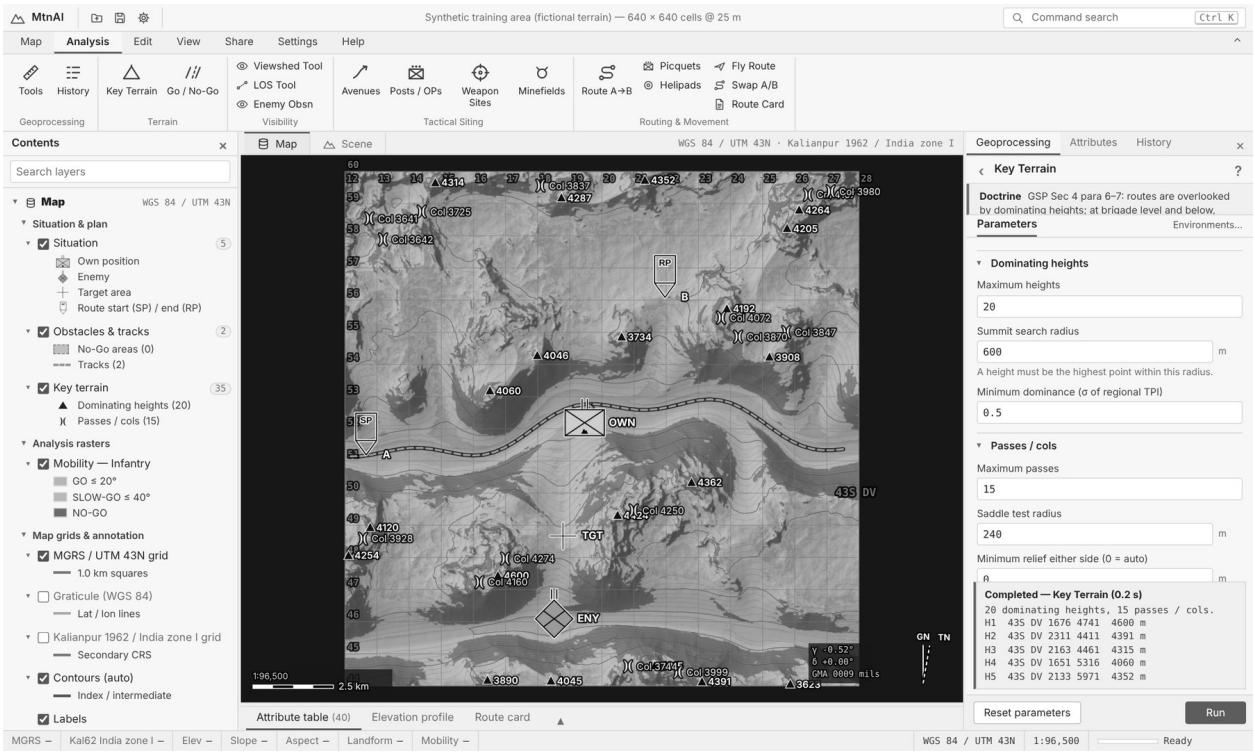


Figure 11. Key terrain: ▲ heights and)(passes.

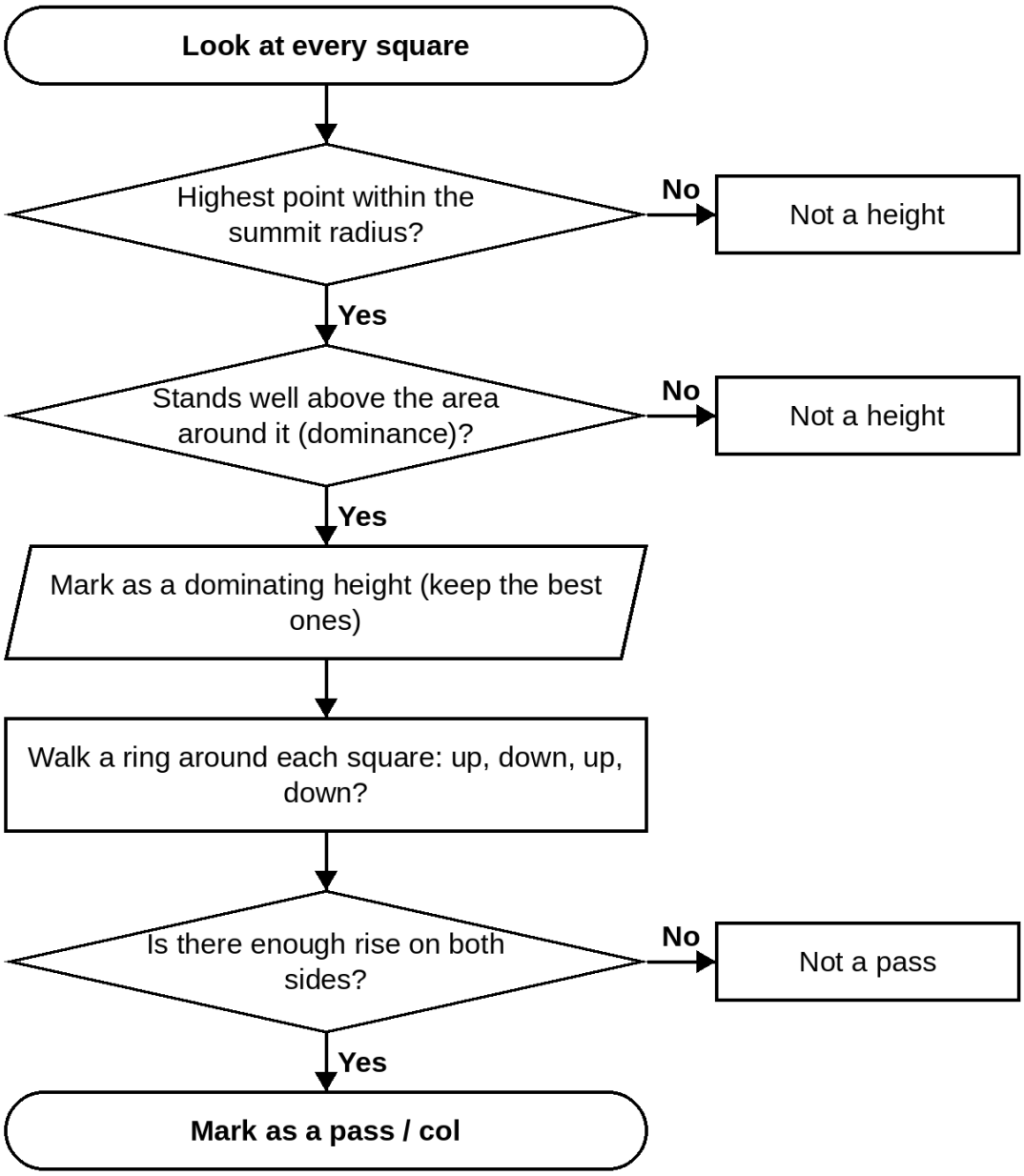


Figure 12. Flowchart — Key Terrain.

Setting	Default	Notes
Dominating heights		
Maximum heights	20	
Summit search radius	600 m	A height must be the highest point within this radius.
Minimum dominance (σ of regional TPI)	0.5	
Passes / cols		
Maximum passes	15	
Saddle test radius	240 m	

Setting	Default	Notes
Minimum relief either side (0 = auto)	0 m	

8. VISIBILITY TOOLS

8.1 Viewshed and Line of Sight

Viewshed: shades all ground an observer can see. **Line of Sight:** answers "can A see B?" and draws the ground profile. Both allow for the earth's curve and the bending of light in air.

Uphill / downhill fire (GSP Section 18). Shots fired steeply up or down strike **high**. Line of Sight gives the angle of sight and tells you to set the sight to **slant range ÷ factor**:

Angle up or down	**0°**	**10°**	**15°**	**20°**	**25°**	**30°**	**45°**	**60°**
Divide range by	1.00	1.02	1.04	1.06	1.10	1.15	1.41	2.00

Doctrine. GSP Sec 4 para 15: heights give long-range observation but rugged ground leaves dead space at mid ranges. A voice in a valley can be heard on ridges 900 m above — cover valleys with listening posts.

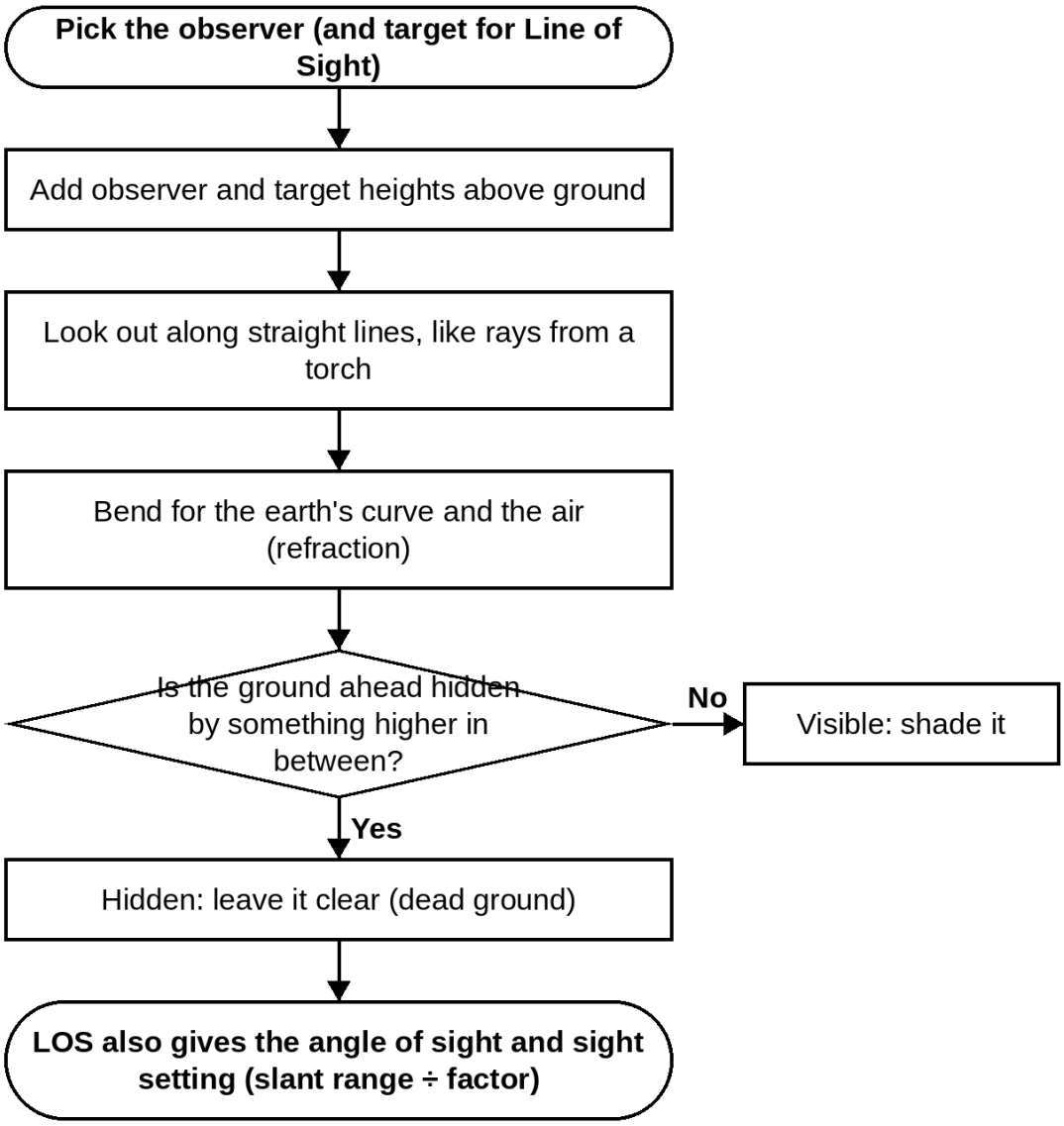


Figure 13. Flowchart — Viewshed and Line of Sight.

Setting	Default	Notes
Observer point	pick on the map	
Observer height above ground	2 m	
Target height above ground	1.5 m	
Radius	5000 m	

Setting	Default	Notes
Observer point	pick on the map	
Target point	pick on the map	
Observer height	2 m	
Target height	2 m	

8.2 Enemy Observation

Guesses where the enemy would put his OPs (the high ground near him) and shades all ground they can see. **Clear ground is hidden from the enemy (defilade)** — good for mortars, guns, helipads and covered routes.

Doctrine. GSP Sec 12: heights are conspicuous — dig down and conceal; use reverse slopes when the forward slope is under enemy observation. Light guns and mortars deploy on reverse slopes and in valleys (Sec 22).

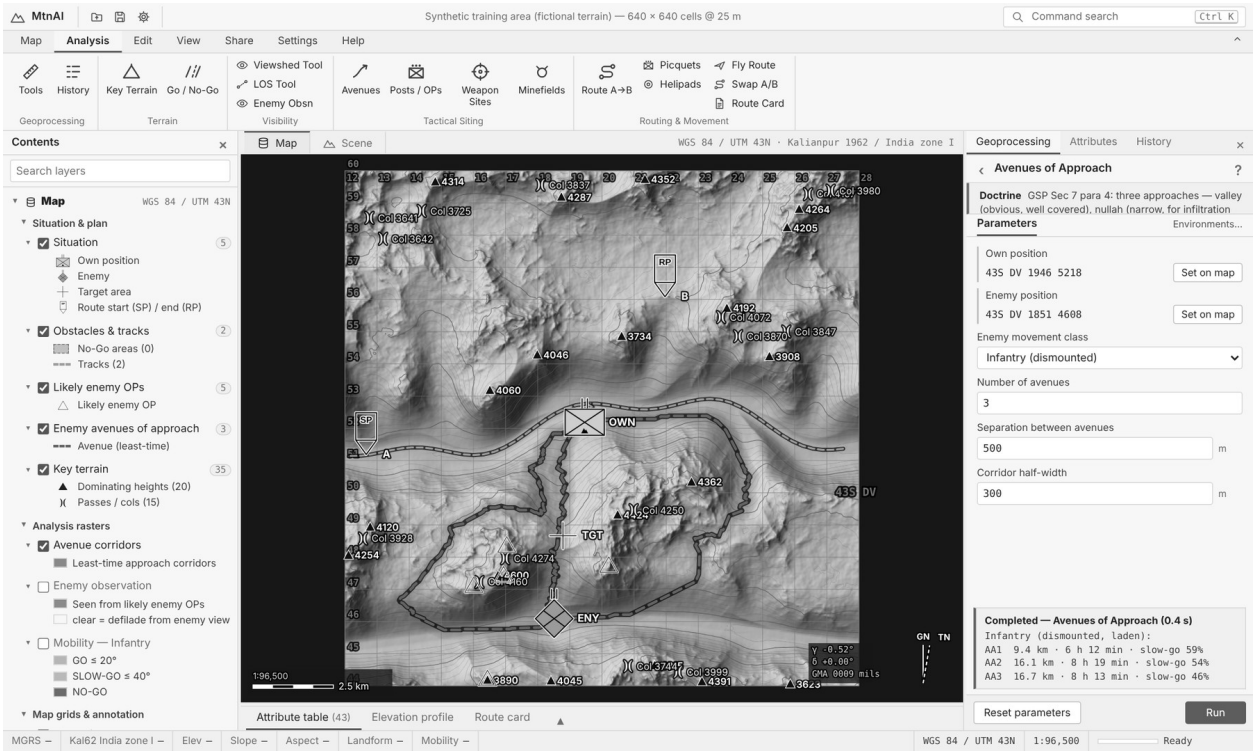


Figure 14. Enemy observation (shaded) and the enemy avenues of approach (dashed).

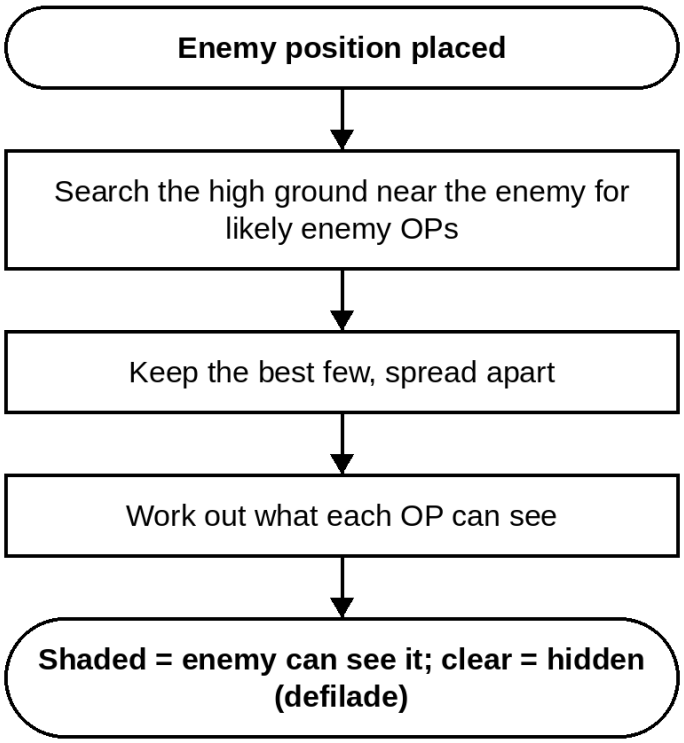


Figure 15. Flowchart — Enemy Observation.

Setting	Default	Notes
Enemy position	set on the map	
Likely enemy OPs to model	5	
OP search radius around enemy	3000 m	
Minimum OP separation	600 m	
Observation range	8000 m	
Enemy observer height	3 m	
Target height (own troops)	1.5 m	
Include the enemy position itself	Yes	

8.3 Avenues of Approach

Finds the quickest ways the enemy can reach your position, for his type of movement, and how long each takes.

Doctrine. GSP Sec 7 para 4: three approaches — valley (obvious, well covered), nullah (narrow, for infiltration and ambush) and ridge line (easy to fight on, hard to maintain). Every conceivable approach must be guarded (Sec 4 para 23).

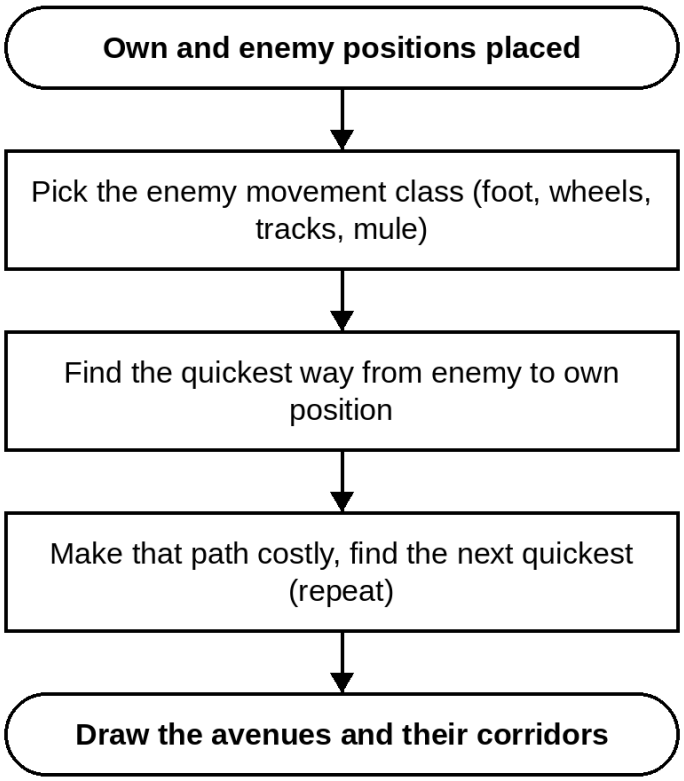


Figure 16. Flowchart — Avenues of Approach.

Setting	Default	Notes
Own position	set on the map	
Enemy position	set on the map	
Enemy movement class	Infantry (dismounted)	
Number of avenues	3	
Separation between avenues	500 m	
Corridor half-width	300 m	

9. SITING TOOLS

9.1 Posts and OPs

Puts posts or OPs on the heights so that, together, they see as much of the enemy's likely ground as possible. It also shows **dead ground** (ground in range that no post sees), which posts can see each other, the **defence tier** of each post and the **strong point**.

Doctrine. GSP Sec 12: occupy dominating features astride and on the flanks of the main axis; posts at platoon level; tiered defence (valley / foothill, mid slope, ridge line); a strong point on the most dominating feature; OPs laterally, in depth and at varying heights.

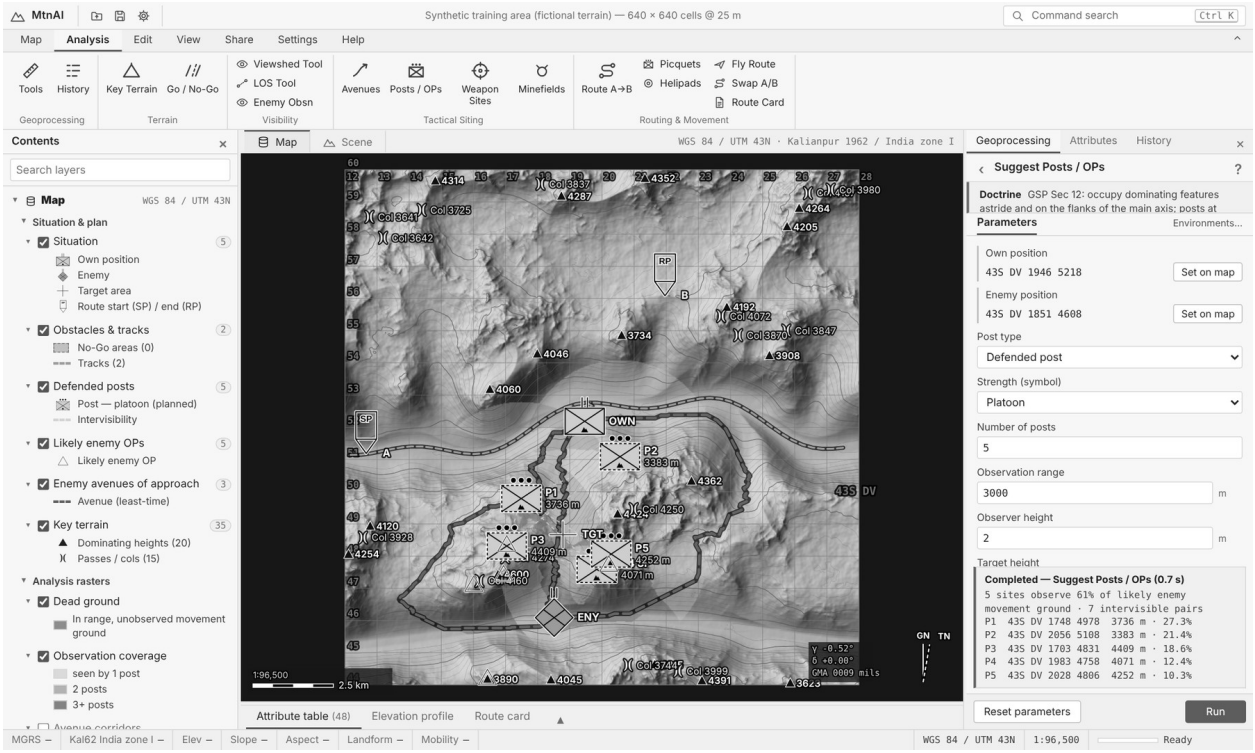


Figure 17. Five suggested posts, their coverage and the result message.

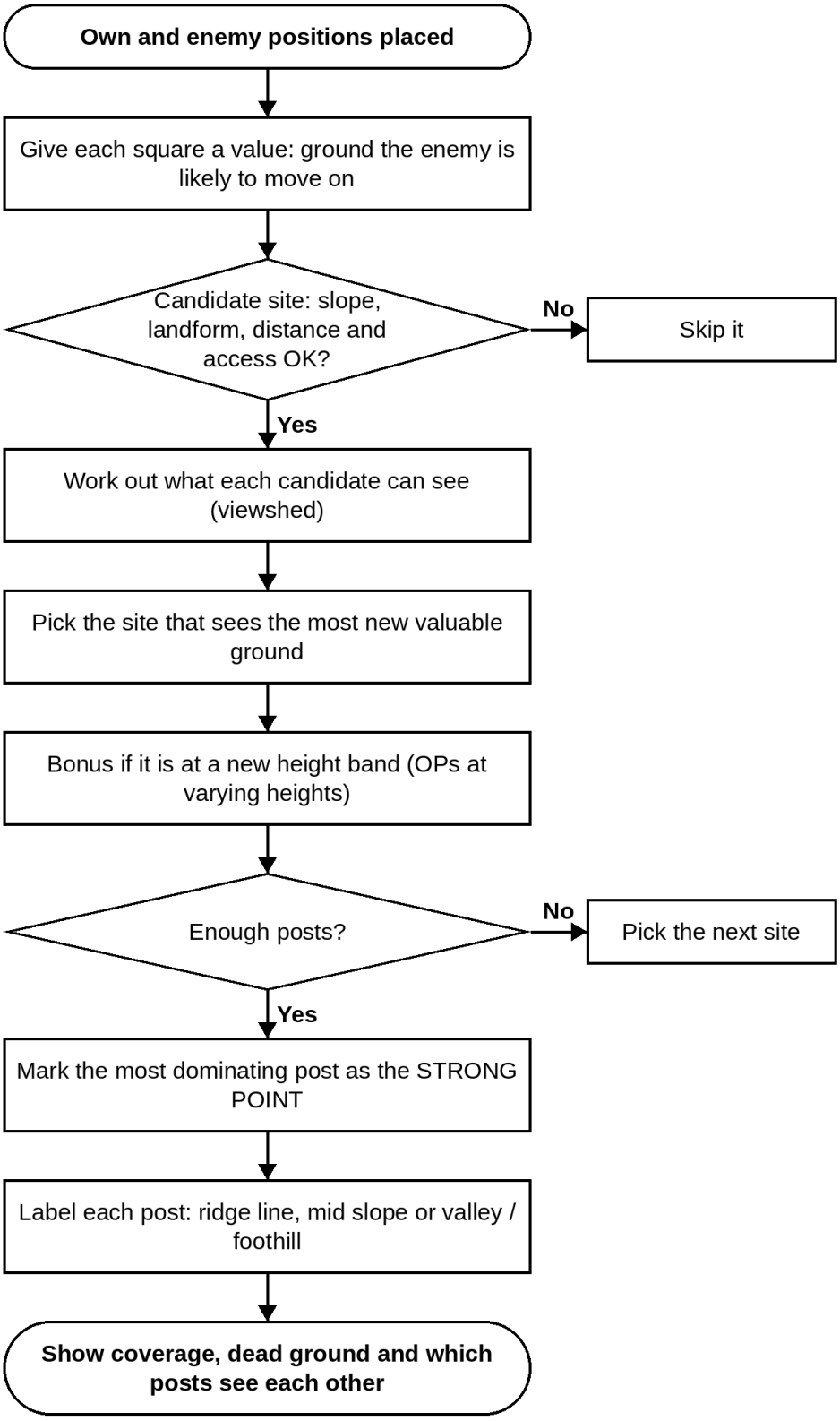


Figure 18. Flowchart — Posts and OPs.

Setting	Default	Notes
Own position	set on the map	
Enemy position	set on the map	
Post type	Defended post	

Setting	Default	Notes
Strength (symbol)	Platoon	
Number of posts	5	
Observation range	3000 m	
Observer height	2 m	
Target height	1.5 m	
Minimum separation	600 m	
Site constraints		
Maximum site slope	35°	
Allowed landforms	Mid slope, Upper slope, Ridge	
Maximum distance from own position	7000 m	
Minimum distance from enemy	500 m	
Must be reachable from own position	Yes	
Access movement class	Infantry (dismounted, laden)	only shown for some choices
Maximum access time (0 = any)	0 h	only shown for some choices
Search & scoring		
Candidate sites evaluated	110	More candidates = better coverage, longer run time.
Value of repeated coverage (0–1)	0.25	0 = no credit for ground already observed; 1 = full credit (mutual support).
Range decay (0–1)	0.5	
Prescore — regional dominance	0.6	
Prescore — local prominence	0.4	
Prescore — absolute height	0.4	
Prescore — closeness to approach	0.4	
Dead-ground value threshold (0–1)	0.25	
Intervisibility check range (× obs range)	2.5	
Height band for OP tiers (0 = off)	150 m	GSP: site OPs laterally, in depth and at varying altitudes so low cloud cannot blind them all.

Setting	Default	Notes
Credit for a new height tier (0–1)	0.15	
Target value — enemy movement ground		
Dismounted GO ground	1	
Dismounted SLOW-GO ground	0.55	
Dismounted NO-GO ground	0.05	
Vehicle GO ground	1	
Vehicle SLOW-GO ground	0.35	
Vehicle NO-GO ground	0	
Share for dismounted movement	0.7	
Share for vehicle movement	0.5	
Bonus on avenues of approach (×)	2	Extra value for ground within the enemy avenue corridors (0 = ignore avenues).
Concentrate on the enemy approach corridor	Yes	
Corridor width (fraction of enemy distance)	0.45	
Minimum corridor width	1500 m	
Value of ground behind own position	0.3	

9.2 Weapon Sites

Picks positions for one weapon at a time. Run it again for each weapon; each gets its own layer.

- **Direct fire** (MG, HMG, RPG-7, 106 RCL, Baktar-Shikan, TOW): sees the target ground in its range band; prefers fire across the enemy's path (enfilade). Ground beyond the mount's **up / down limits** is dead to the weapon. Each site shows its arc of fire and **sight setting**.
- **Indirect fire** (60 / 81 / 120 mm mortars, 105 / 122 / 130 / 155 mm guns): behind own troops, in range of the target, able to **clear the crests**, ideally hidden from enemy OPs.
- **Air defence** (Anza, 35 mm): covers the low flying routes along valleys and defiles from dominating ground.

Doctrine. GSP Sec 18: fire up- or downhill strikes high — set the sight to slant range ÷ factor (10° 1.02, 20° 1.06, 25° 1.10, 60° 2.0). Plunging fire leaves dead ground; cover it with 60 / 81 mm mortars, riflemen and mines. Air defence covers defiles, gun areas and logistic sites from dominating features (Sec 22).

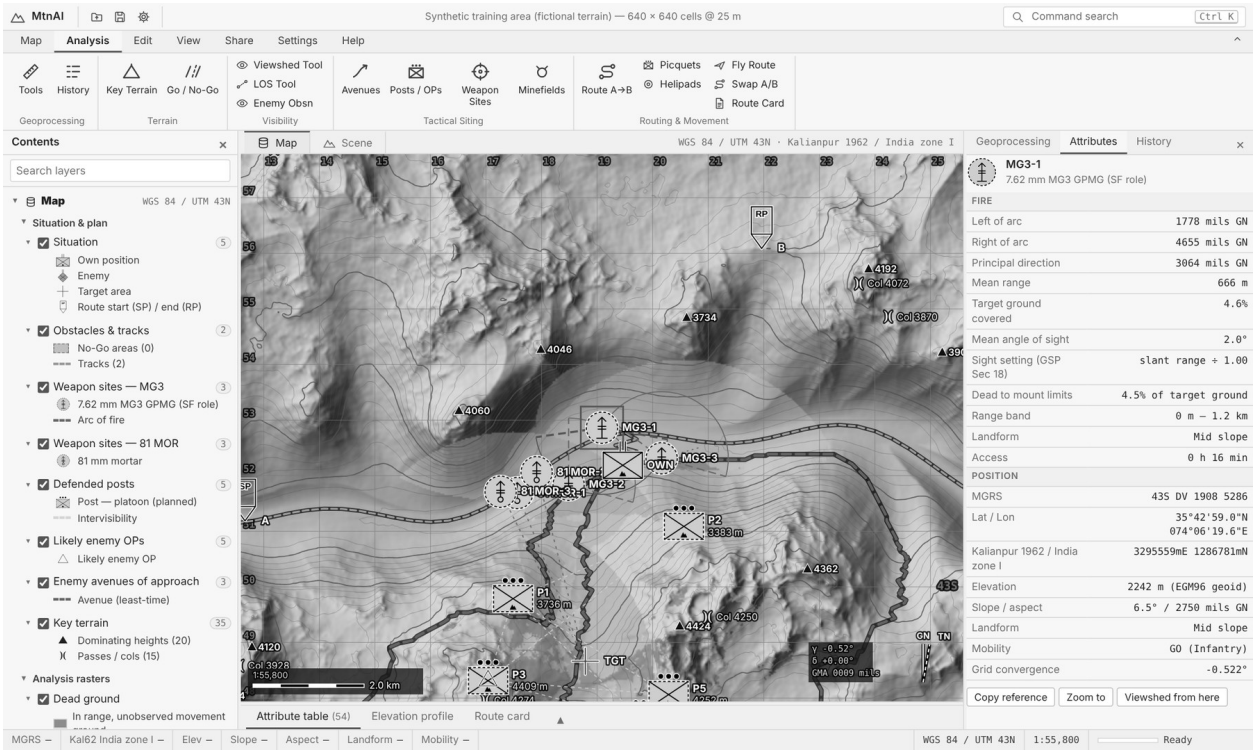


Figure 19. An MG3 site: arc of fire, mean angle of sight, sight setting and the share of ground dead to the mount limits.

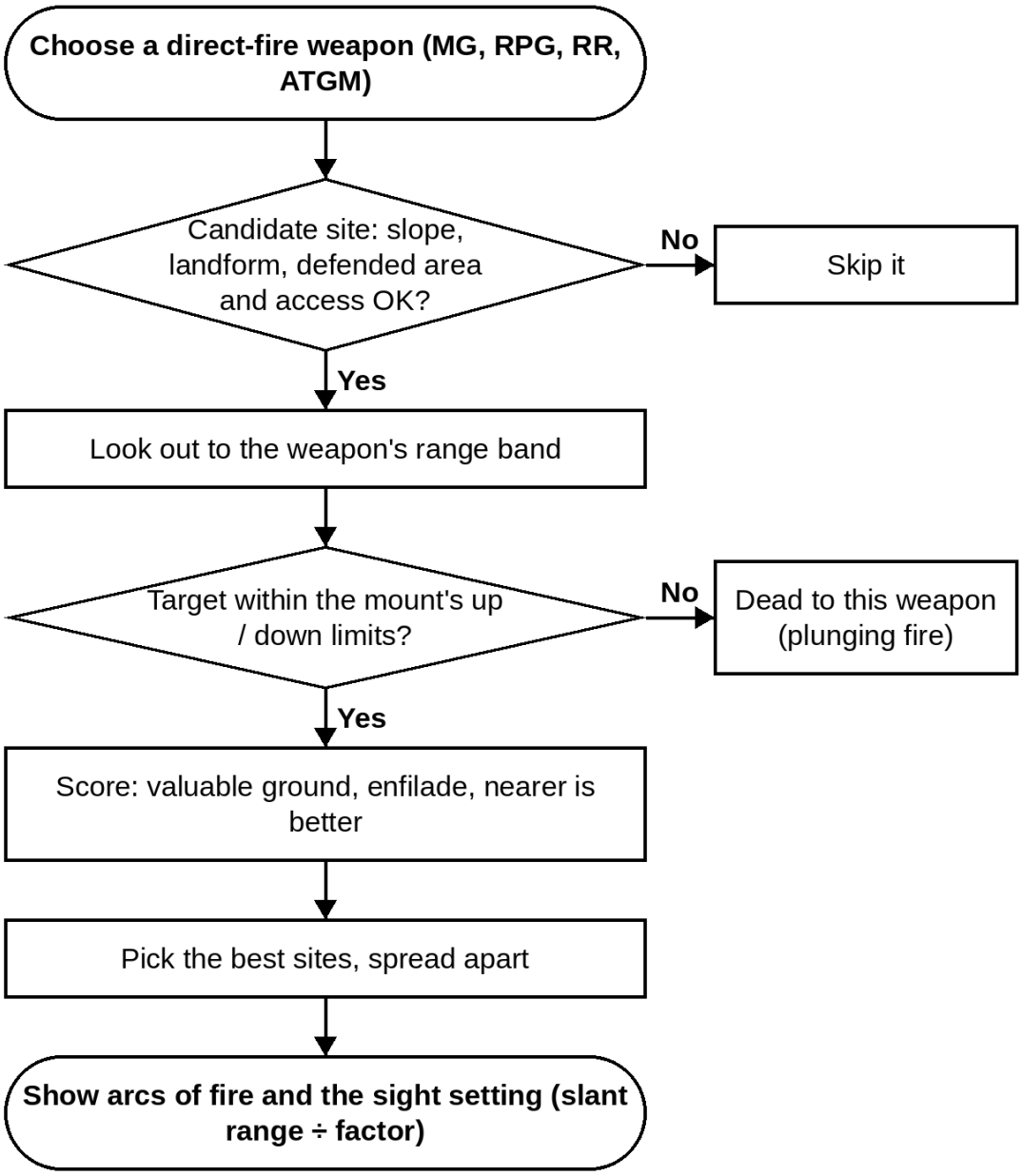


Figure 20. Flowchart — direct-fire weapon sites.

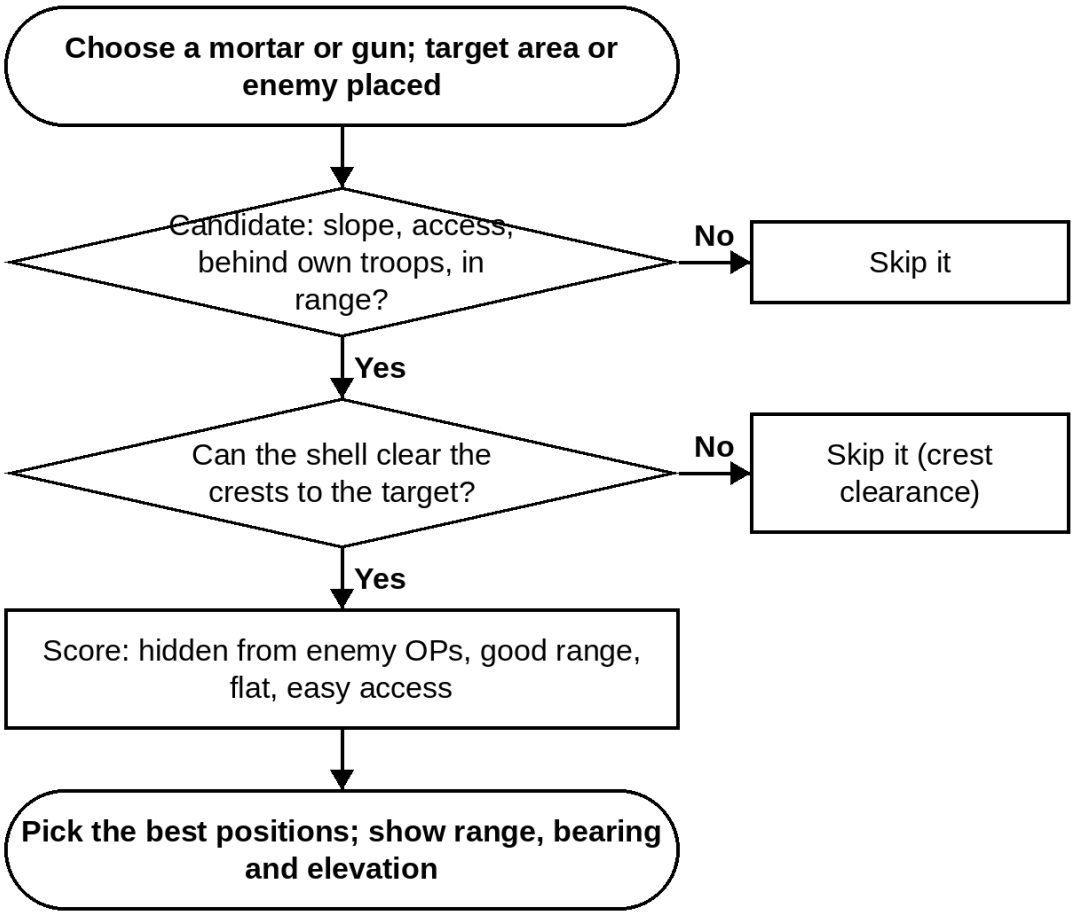


Figure 21. Flowchart — indirect-fire (mortar and gun) positions.

Setting	**Default**	**Notes**
Own position	set on the map	
Enemy position	set on the map	
Target area	set on the map	
Weapon	7.62 mm MG3 GPMG (SF role) — Direct fire — anti-personnel	
Number of sites	3	
Weapon data (saved to Settings ▸ Weapon data)		
Minimum range	from weapon data	
Maximum / effective range	from weapon data	
Maximum site slope	from weapon data	
Sight height	from weapon data	
Target height	from weapon data	

Setting	Default	Notes
Mount elevation limit (blank = none)	from weapon data	GSP Sec 18: MG on mount +6° / −8.5°; 75 mm RR +30° / −5°. Ground beyond the limits is dead to the weapon (plunging fire).
Mount depression limit (blank = none)	from weapon data	
Access movement class	from weapon data	
Site constraints		
Defended-area radius around own position	3000 m	
Minimum separation	300 m	
Must be reachable from own position	Yes	
Maximum access time (0 = any)	0 h	
Enfilade preference (0–1, blank = weapon default)	—	Credit for fire across the enemy approach rather than head-on.
Position must be hidden from enemy observation	No	indirect fire only
Maximum forward of own position (fraction)	0.25	indirect fire only
Minimum range to target (× max range)	0.15	indirect fire only
Maximum range to target (× max range)	0.92	indirect fire only
Preferred range to target (× max range)	0.55	indirect fire only
Target-area spread (crest clearance fan)	400 m	indirect fire only
Crest clearance margin	15 m	indirect fire only
Search & scoring		
Candidate sites evaluated	120	
Value of overlapping arcs (0–1)	0.5	
Range decay (0–1)	0.35	
Landform preference	0.25, 0.5, 0.6, 0.85, 1, 0.55	
Landform preference (site)	0.1, 0.3, 0.45, 0.7, 1, 1	air defence only

Setting	Default	Notes
Air corridor value by landform	1, 0.8, 0.7, 0.25, 0.1, 0.05	Where low-flying aircraft are expected (helicopters follow valleys). air defence only
Landform preference	0.7, 0.9, 1, 0.6, 0.35, 0.15	indirect fire only
Weight — crest clearance	0.35	indirect fire only
Weight — defilade from enemy view	0.3	indirect fire only
Weight — range suitability	0.15	indirect fire only
Weight — flat ground	0.1	indirect fire only
Weight — access time	0.1	indirect fire only
Target value — enemy movement ground		
Dismounted GO ground	1	
Dismounted SLOW-GO ground	0.55	
Dismounted NO-GO ground	0.05	
Vehicle GO ground	1	
Vehicle SLOW-GO ground	0.35	
Vehicle NO-GO ground	0	
Bonus on avenues of approach (×)	2	Extra value for ground within the enemy avenue corridors (0 = ignore avenues).
Concentrate on the enemy approach corridor	Yes	
Corridor width (fraction of enemy distance)	0.45	
Minimum corridor width	1500 m	
Value of ground behind own position	0.3	

Weapon	Group	Range (m)	Site slope	Mount limits
7.62 mm MG3 GPMG (SF role)	Direct fire — anti-personnel	0–1200	30°	+6° / −8.5°
12.7 mm HMG (Type 54 / W-85)	Direct fire — anti-personnel	0–1600	25°	—
RPG-7 (AT rocket launcher)	Direct fire — anti-armour	30–300	35°	—
106 mm RCL (M40)	Direct fire — anti-armour	50–1100	12°	+30° / −5°

Weapon	Group	Range (m)	Site slope	Mount limits
Baktar-Shikan ATGM	Direct fire — anti-armour	100–3000	20°	—
TOW-2A ATGM	Direct fire — anti-armour	65–3750	15°	—
60 mm light mortar	Indirect fire — mortars	100–2500	15°	—
81 mm mortar	Indirect fire — mortars	100–5600	12°	—
120 mm mortar	Indirect fire — mortars	450–7200	10°	—
105 mm M56 pack howitzer (mountain arty)	Indirect fire — artillery	1000–10500	10°	—
122 mm Type 54-1 howitzer	Indirect fire — artillery	2000–11800	8°	—
130 mm Type 59-1 field gun	Indirect fire — artillery	3000–27000	6°	—
155 mm M198 howitzer	Indirect fire — artillery	2500–22400	6°	—
Anza Mk-II MANPADS	Air defence	500–5000	35°	—
Anza Mk-III MANPADS	Air defence	500–6000	35°	—
35 mm Oerlikon GDF twin AD gun	Air defence	0–4000	8°	—

All weapon figures are open-source planning values. Change them in **Settings ▸ Weapon data** or in the tool.

9.3 Minefields

Finds the **narrowest gaps (defiles)** on the enemy's avenues and draws an AT or AP minefield across each, tied into No-Go ground. It counts mines and says if each field is watched by your posts or weapons.

Doctrine. GSP Sec 12: gaps between posts are covered by mines, fire and obstacles; density thickest in valleys and foothills. Mines lose effect in heavy snow and rocky ground and wash away on slopes — use wire there.

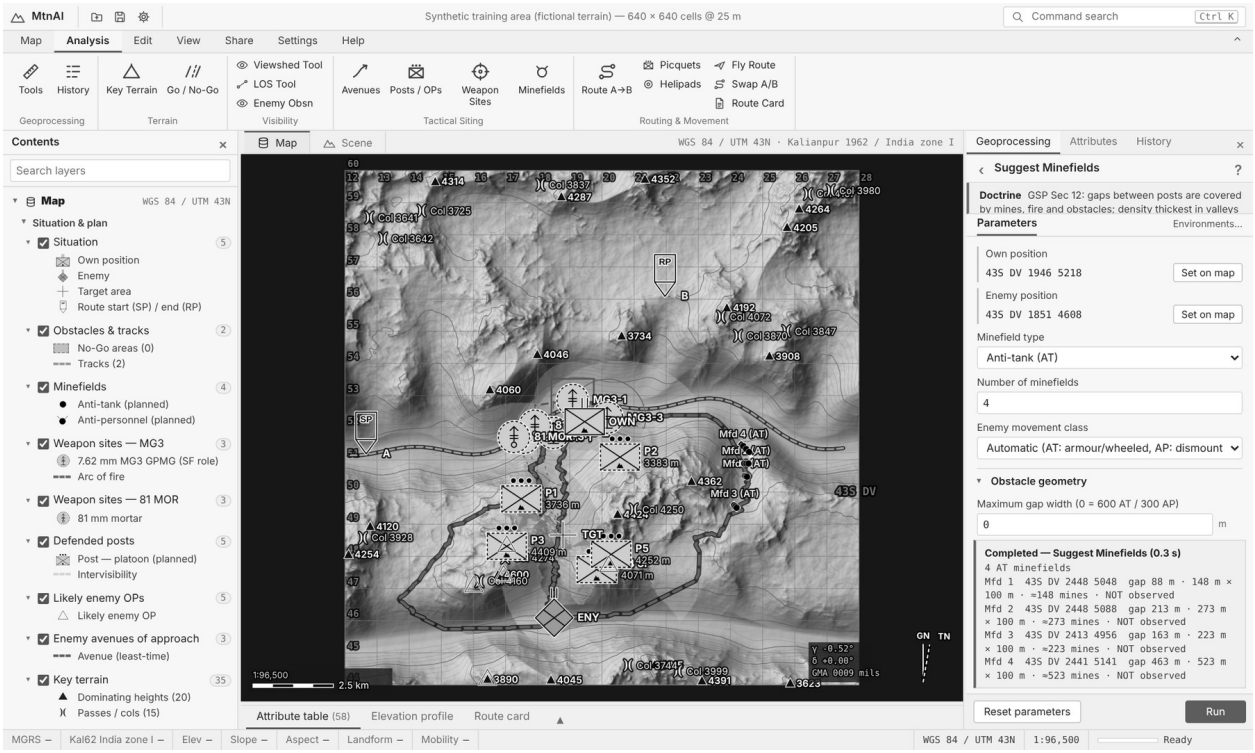


Figure 22. Suggested anti-tank minefields across the enemy avenues.

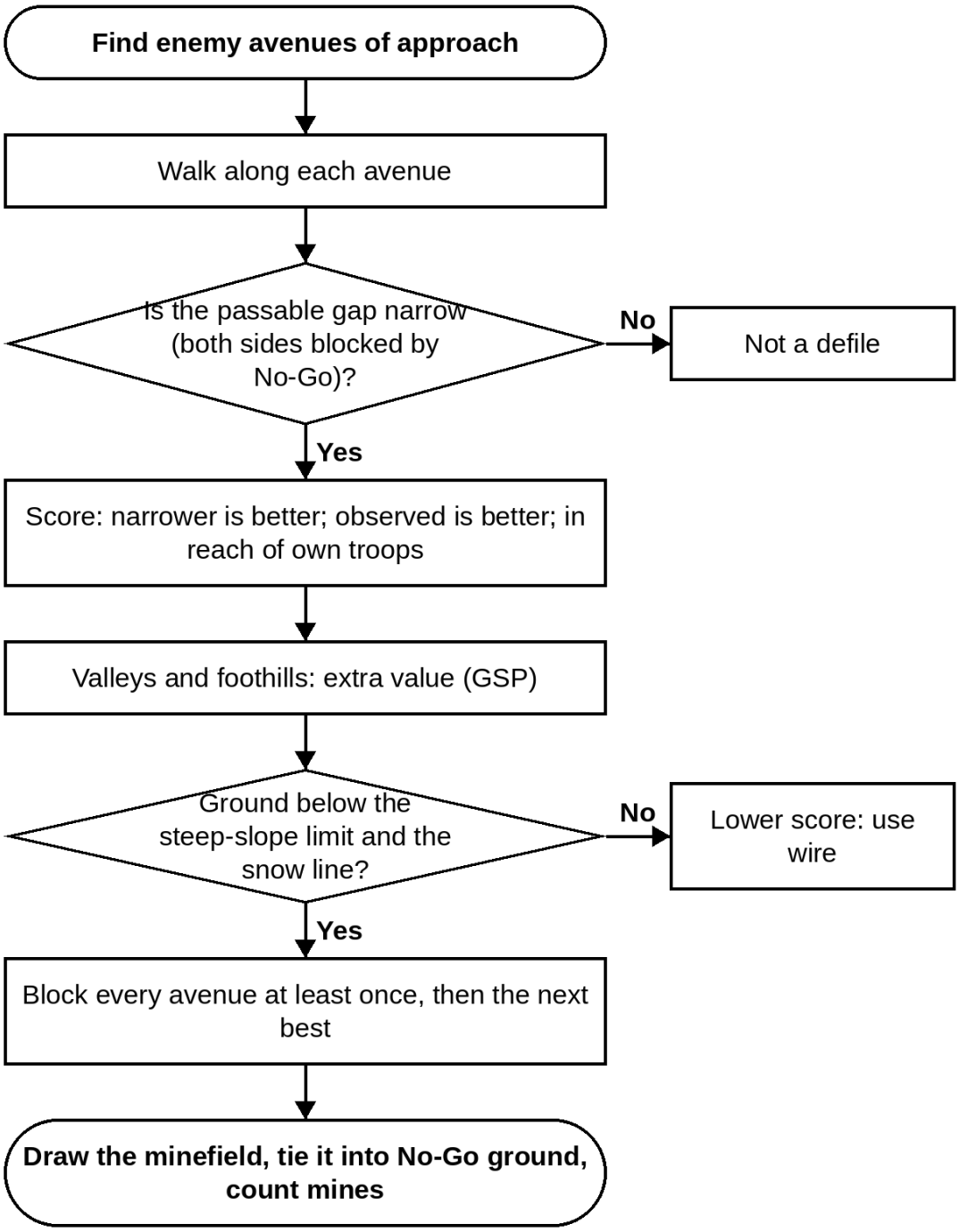


Figure 23. Flowchart — Minefields.

Setting	Default	Notes
Own position	set on the map	
Enemy position	set on the map	
Minefield type	Anti-tank (AT)	
Number of minefields	4	
Enemy movement class	Automatic (AT: armour/wheeled, AP: dismounted)	

Setting	Default	Notes
Obstacle geometry		
Maximum gap width (0 = 600 AT / 300 AP)	0 m	
Depth (0 = 100 AT / 40 AP)	0 m	
Density (0 = 1 AT / 3 AP)	0 mines/m	
Tie-in into NO-GO ground	30 m	
Siting constraints		
Minimum distance from own position	300 m	
Maximum distance from own position	3000 m	
Minimum separation	400 m	
Avenues of approach considered	3	
Block every avenue at least once	Yes	
Only sites under own observation / fire	No	
Search & scoring		
Orientations tested at each defile	8	
Weight of narrowness (0–1)	0.7	
Factor if not observed (0–1)	0.45	
Factor behind own position (0–1)	0.2	
Factor outside distance band (0–1)	0.15	
Ground (GSP Mountain Warfare)		
Extra value in valleys / foothills	0.25	GSP: minefield density is thickest in the valleys and foothills.
Steep-slope limit	30°	GSP: mines on slopes wash away in rain and slides; use wire on rocky / steep slopes.
Factor on steeper ground (0–1)	0.35	
Snow line (0 = off)	0 m	GSP: heavy snow cover greatly reduces the effect of mines.
Factor above the snow line (0–1)	0.4	

9.4 Helipad Sites (HLS)

Finds flat, hidden places where a helicopter can land, near your position.

Doctrine. GSP Sec 22 para 19: sites for repeated use ≤ 4° slope; in emergency 4–10° for observation / utility and 4–20° for cargo helicopters. Shield the site and its approach from enemy observation; avoid dust (dry barren ground) — hard ground, rock or grass is best.

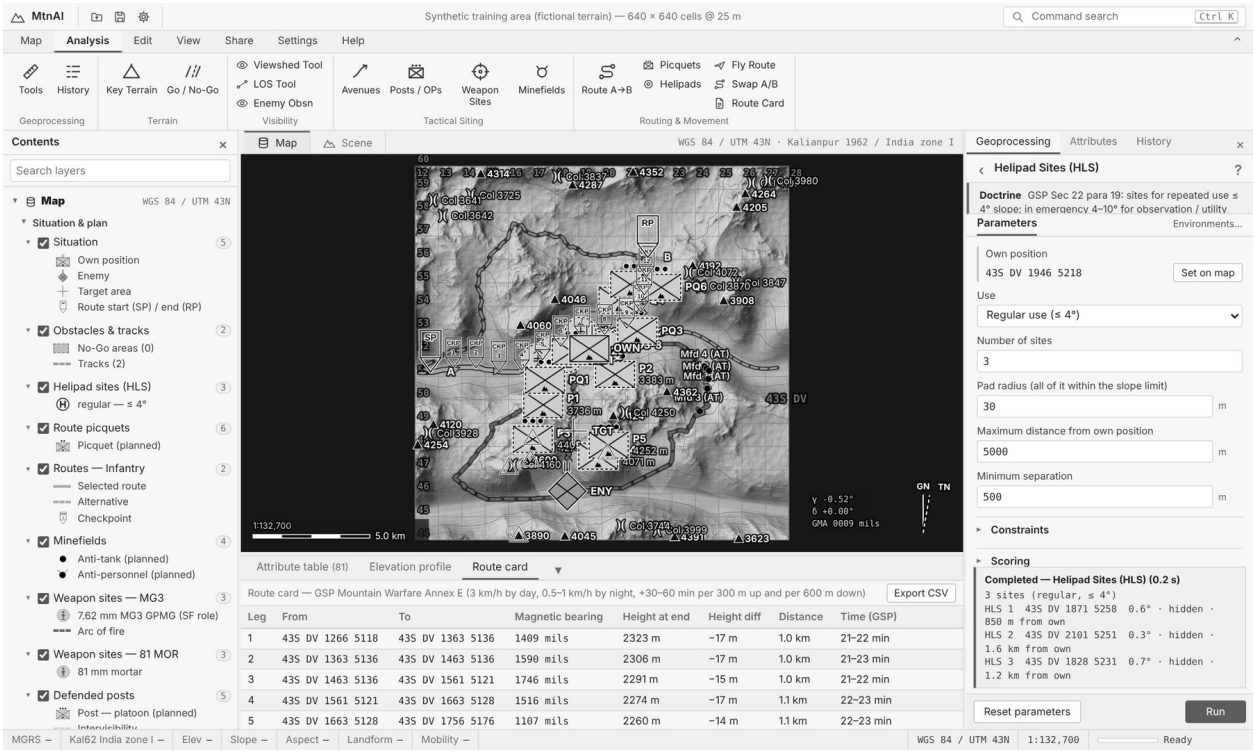


Figure 24. Helipad sites (H) and their details.

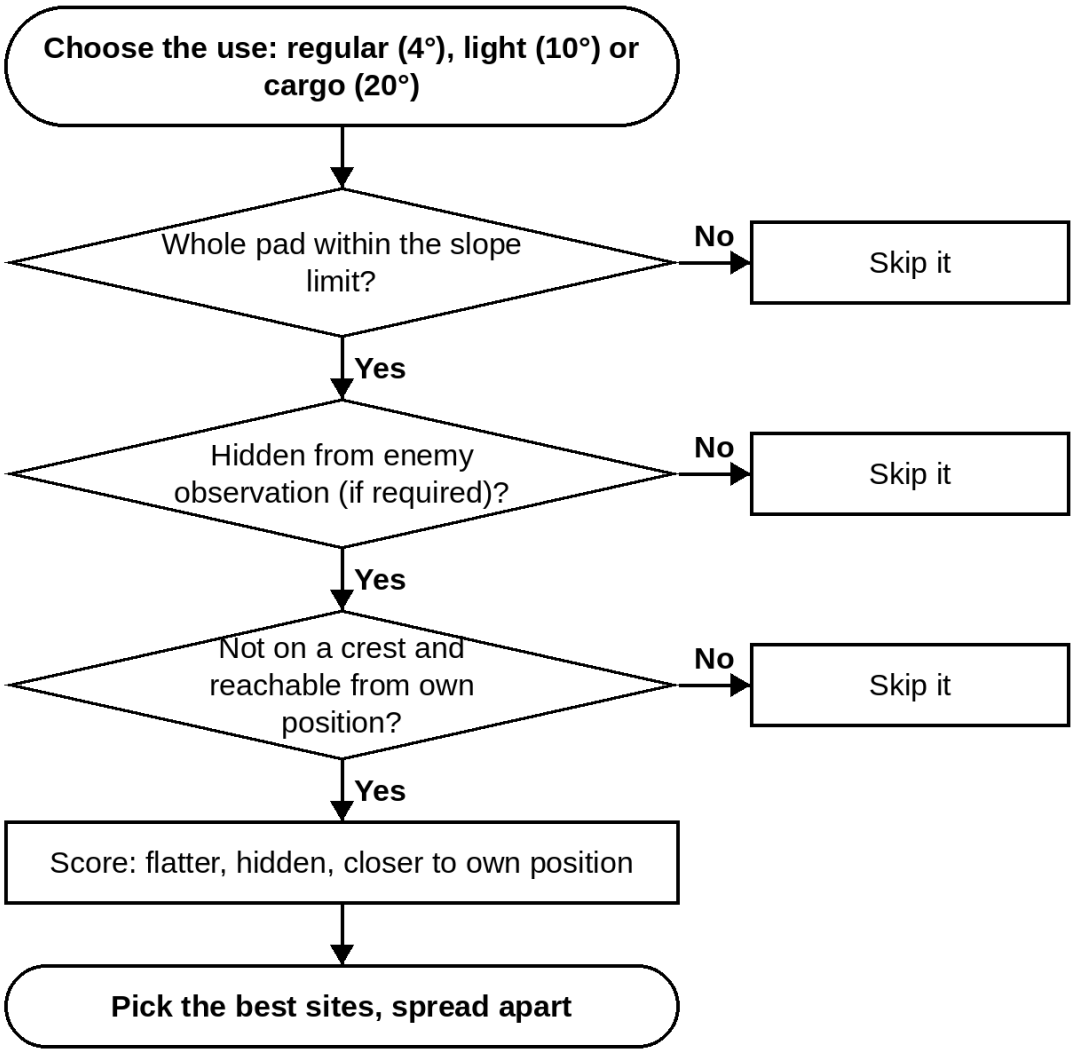


Figure 25. Flowchart — Helipad Sites.

Setting	**Default**	**Notes**
Own position	set on the map	
Use	Regular use (≤ 4°)	
Number of sites	3	
Pad radius (all of it within the slope limit)	30 m	
Maximum distance from own position	5000 m	
Minimum separation	500 m	
Constraints		
Hidden from enemy observation	Yes	
Not on crests (skyline, wind)	Yes	
Reachable from own position	Yes	

Setting	Default	Notes
Access movement class	Infantry (dismounted, laden)	only shown for some choices
Maximum access time (0 = any)	0 h	only shown for some choices
Scoring		
Weight — flatness	0.4	
Weight — concealment	0.35	
Weight — closeness to own position	0.25	

10. MOVEMENT TOOLS

10.1 Least-Time Route (A to B) and the Route Card

Finds the quickest safe path between Start A and End B for the chosen movement class. It knows that walking uphill is slow, steep downhill is slow too, and NO-GO ground is impossible. It can also keep you **out of enemy view**, **off skylines** and **out of nullah beds**.

Each route shows: length, climb and descent, steepest part, share in SLOW-GO and in enemy view, the model time and the **GSP time** (day and night). The **Route Card** (bottom dock) lists every leg as in GSP Annex E: start and end reference, magnetic bearing, height at end, height difference, distance and time. Export it as CSV.

Doctrine. GSP Sec 4 / 7 / Annex E: measure distance in time — 3 km/h by day, 0.5–1 km/h by night, plus 30–60 min per 300 m of ascent and per 600 m of descent. Avoid losing height (a 3 km ridge detour beats a 600 m down-and-up). Follow ridges, not gullies. Prepare a route card.

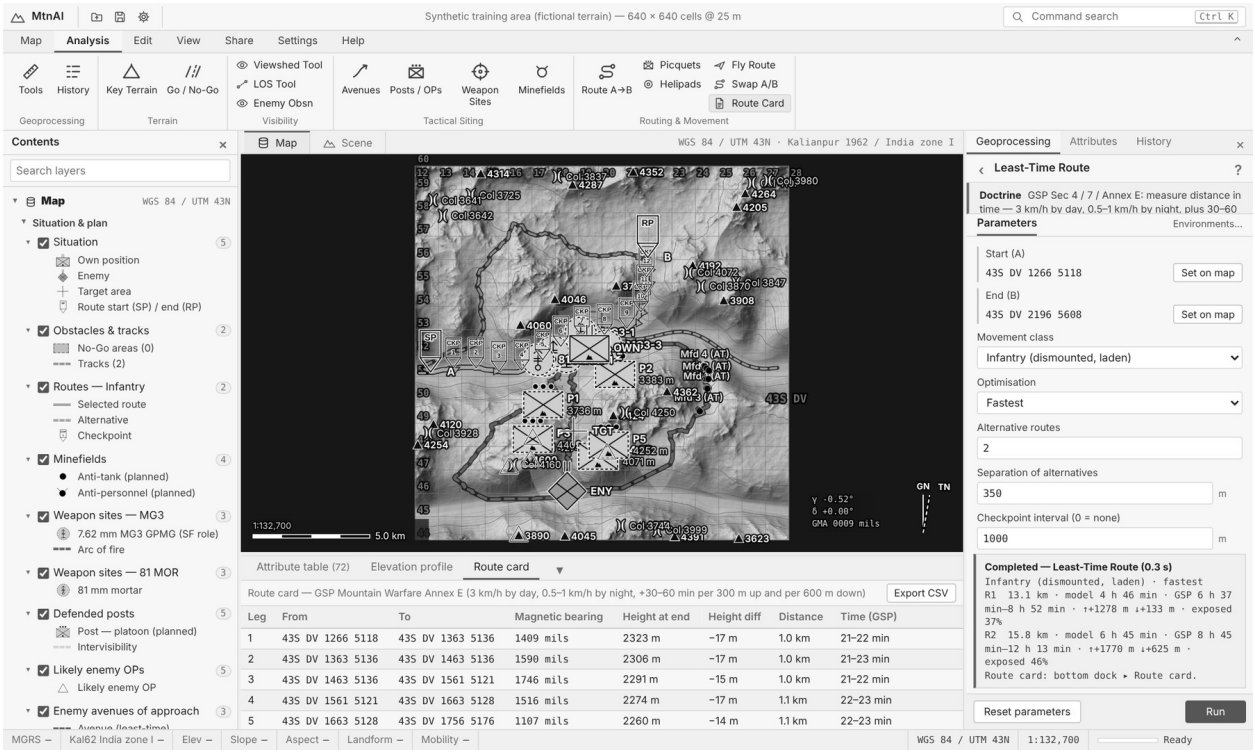


Figure 26. Route A→B with checkpoints and the GSP route card in the bottom dock.

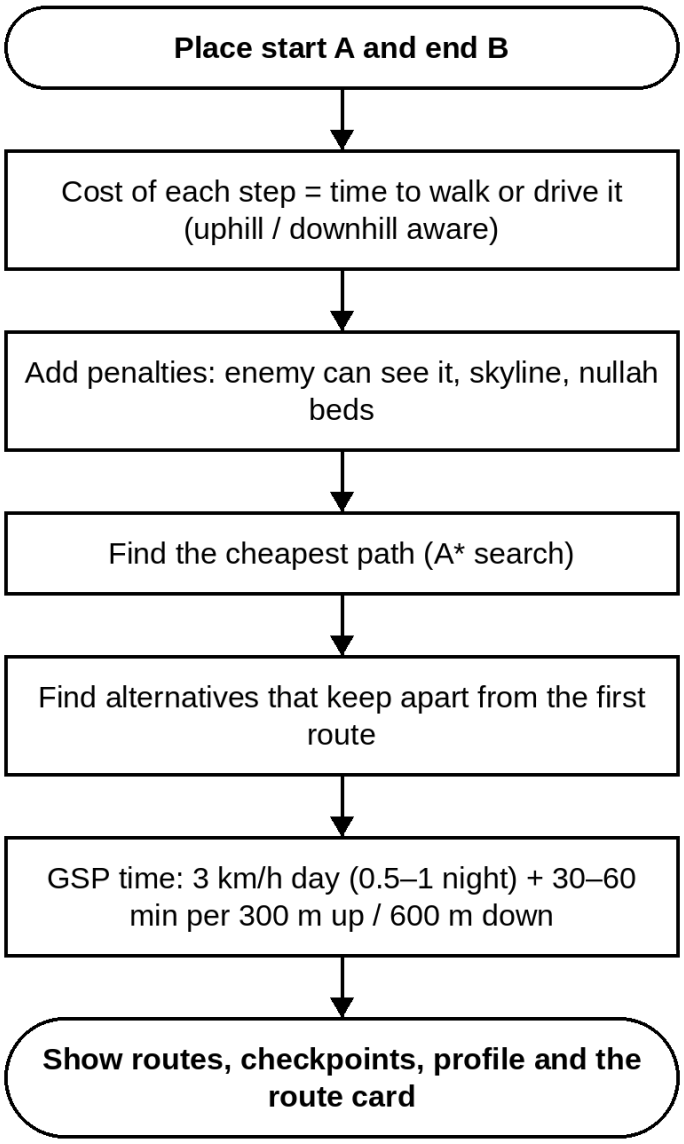


Figure 27. Flowchart — Least-Time Route.

Setting	Default	Notes
Start A	set on the map	
End B	set on the map	
Movement class	Infantry (dismounted, laden)	
Optimisation	Fastest	
Exposure penalty	3 ×	only shown for some choices
Skyline (crest) penalty	0.8 ×	only shown for some choices
Alternative routes	2	
Separation of alternatives	350 m	
Checkpoint interval (0 = none)	1000 m	

Setting	Default	Notes
Keep out of nullah beds (penalty)	0.5 ×	GSP: routes follow ridge lines rather than gullies and stream beds (scree, vertical patches, ambush). Tracks are exempt.
Route card (GSP Annex E)		
Leg length	1000 m	
Night move (0.5–1 km/h instead of 3 km/h)	No	

10.2 Route Picquets

Puts picquets on the heights that overlook the chosen route, so the column below is safe ("crowning the heights").

Doctrine. GSP Sec 6: the axis passes through valleys and defiles dominated by flanking heights — secure (picquet) these heights for the main body and to protect the main supply route.

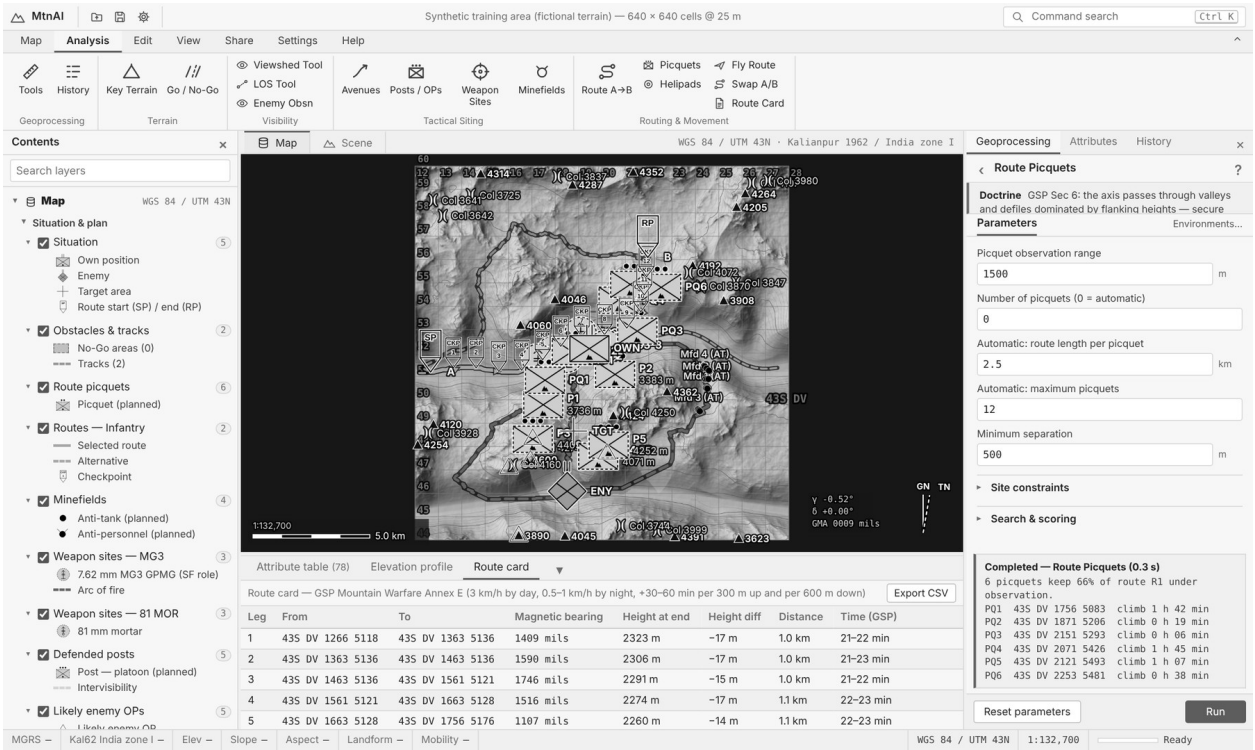


Figure 28. Route picquets on the heights above the route.

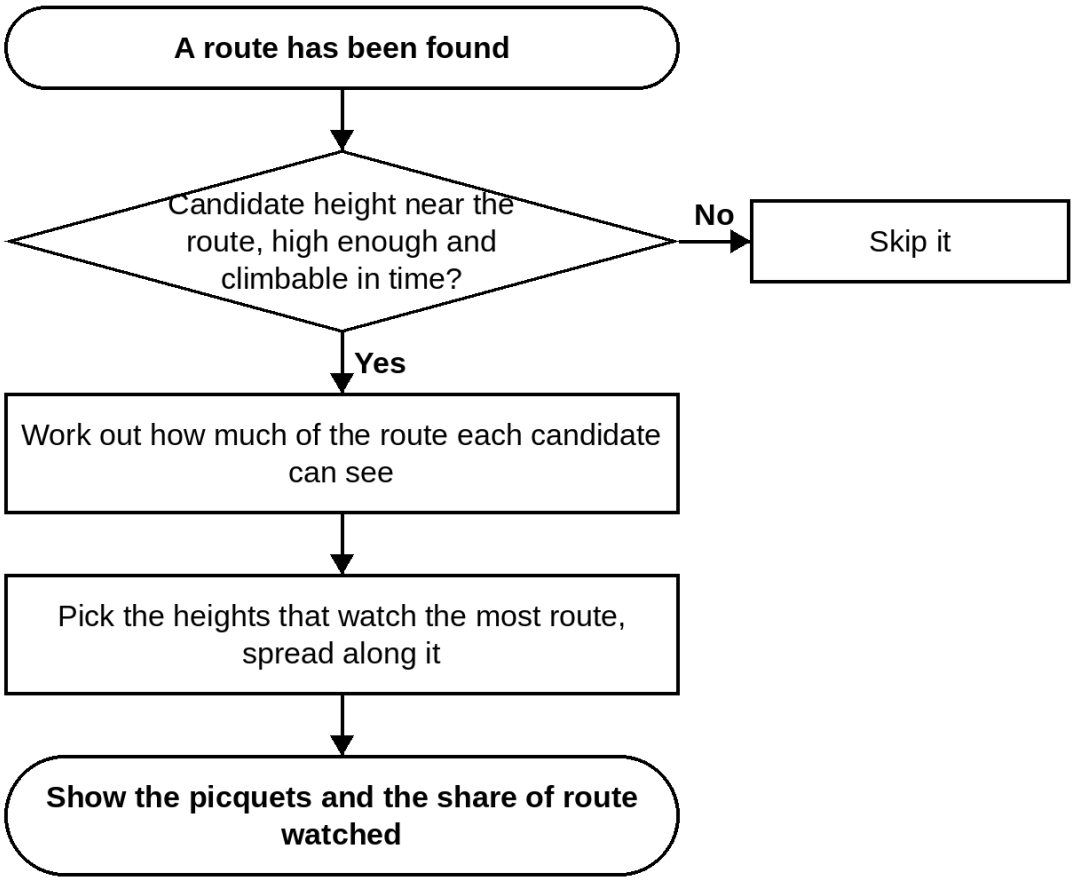


Figure 29. Flowchart — Route Picquets.

Setting	Default	Notes
Picquet observation range	1500 m	
Number of picquets (0 = automatic)	0	
Automatic: route length per picquet	2.5 km	only shown for some choices
Automatic: maximum picquets	12	only shown for some choices
Minimum separation	500 m	
Site constraints		
Maximum distance from route	2000 m	
Lowest landform allowed	Upper slope	
Maximum site slope	38°	
Maximum climb time from route	4 h	
Search & scoring		
Candidate sites evaluated	120	
Value of overlapping coverage (0–1)	0.15	

Setting	Default	Notes
Observer height	2 m	
Target height	1.5 m	

10.3 Fly Route (3D)

Flies the camera along the route in the Scene view. Press Esc to stop.

Setting	Default	Notes
Speed	260 m/s	
Height above route	140 m	

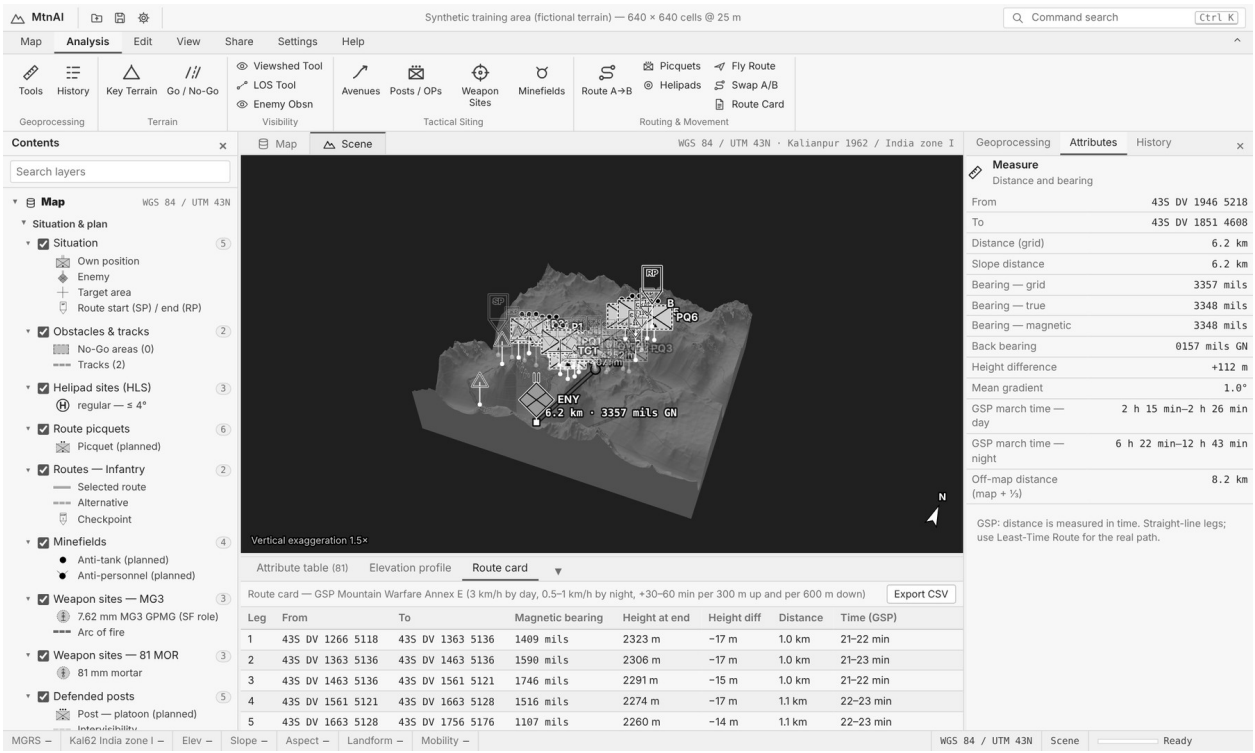


Figure 30. The Scene (3D) view.

11. LAYERS, TABLES AND SAVING

Contents pane

- Tick or untick a layer to show or hide it. Click ▸ to open its key.
- Right-click: **Zoom To Layer**, **Attribute Table**, **Transparency**, **Remove**, **Symbology settings**, **Properties**.
- Double-click the height layer to change the base map (shaded relief, height colours, slope, aspect, landforms, ruggedness) and see its statistics.

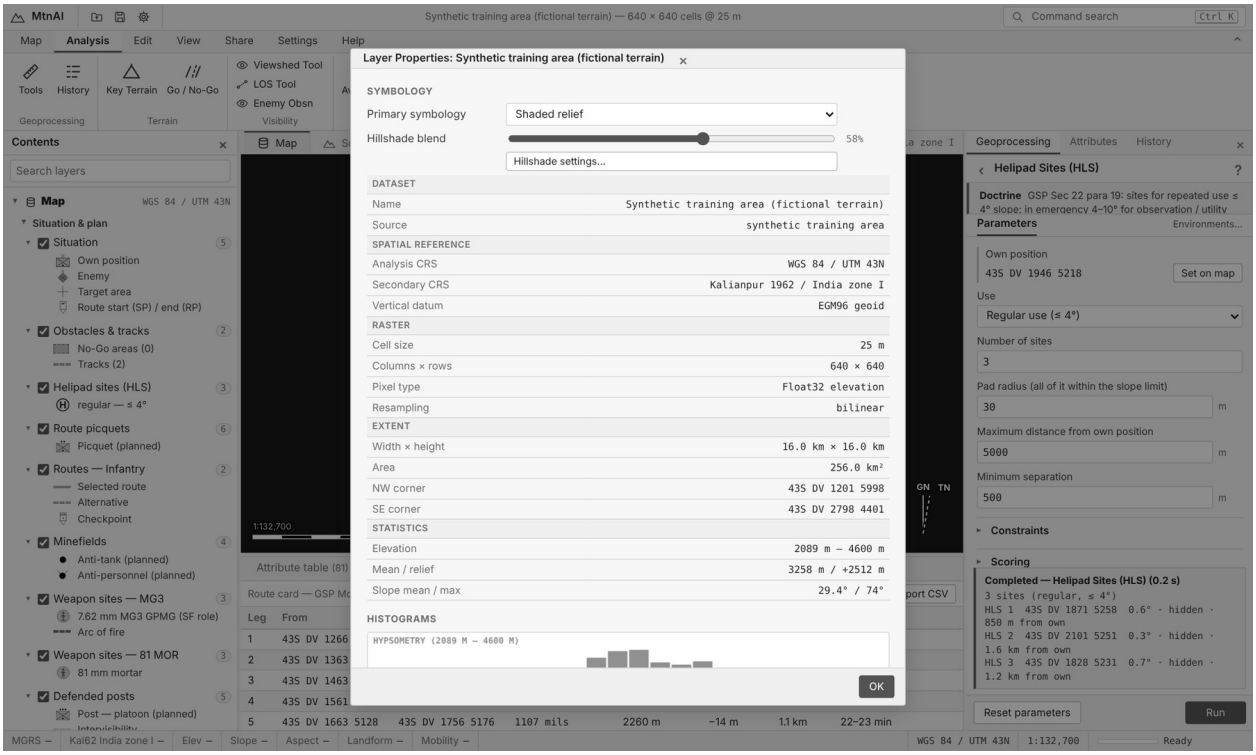


Figure 31. Layer Properties of the DEM: symbology, data source, coordinate system, extent and statistics.

Attribute table and Attributes pane

- **Attribute table** (Ctrl+T): every planned item, one row each. Choose a layer, filter, sort by clicking a column, click a row to zoom. **Export CSV** saves it.
- **Attributes** pane: everything about the item you clicked, with its position in every coordinate system. Buttons: Copy reference, Zoom to, Viewshed from here.

Saving and sharing (Share tab)

Button	Gives you
Save Plan / Open Plan	A small .json file with your markers, drawings and results. Open it again later on the same DEM.
GeoJSON	All features with NATO symbol codes (SIDC) for other GIS or C2 systems.
Table (CSV)	The attribute table for Excel.
Raster	Any layer as GeoTIFF or ASCII grid.
View (PNG)	A picture of the map or the 3D view.

Button	Gives you
Report	A printable terrain assessment with maps and every result (Ctrl+P).

12. SETTINGS

Settings (Ctrl+,) are grouped. Changed settings are marked; you can search, export, import and restore defaults.

Group	What you can change
Coordinate systems	Analysis UTM zone and hemisphere; second system (Kalianpur etc.) or a custom one.
Grid references	Main format, number of figures, spacing, lat / long decimals.
North reference	Grid, true or magnetic; declination and yearly change; mils, degrees or gradians.
Units	Distance, height, area and time units; vertical datum.
Map grids	Which grids, their spacing, labels and colours.
Map display	Base map, hillshade sun direction, contours.
Symbology	NATO symbol size, colours, labels.
Overlays	Colours and transparency of the analysis layers.
3D view	Height exaggeration, mesh detail, light, haze, fly speed.
Analysis	Earth curve and refraction, observer heights, landform scales, altitude slow-down, route options.
Movement classes	GO / SLOW-GO slopes and speeds for each class.
Weapon data	Range, site slope, heights, access and mount limits of each weapon.
Data, Report, Interface	Import options, report header, theme (dark, light, high contrast, night red), text size.

13. DOCTRINE IN MTNAI (GSP MOUNTAIN WARFARE)

These rules from the pamphlet are built into the tools. Help ▸ **Doctrine** shows the same table inside the app.

Subject	GSP teaching	Where in MtnAI
Planning (Sec 4)	Measure distance in time. Map distance + ⅓ ≈ ground distance. Avoid losing height (a 3 km ridge detour beats a 600 m down-and-up). Follow ridges, not gullies. Make a route card.	Route: GSP time, nullah penalty, route card. Measure: GSP time.
Rate of march (Sec 7, Annex E)	Day 3 km/h; night 0.5–1 km/h; add 30–60 min for every 300 m up and every 600 m down.	GSP time on every route and route-card leg.
Objectives (Sec 4, 7)	Seize or hold the dominating heights; objectives on the flanks and rear that dominate the enemy.	Key Terrain.
Approaches (Sec 7)	Valley (obvious, well covered), nullah (infiltration, ambush), ridge line (easy to fight on, hard to supply). Guard every approach.	Avenues of Approach.
Defence (Sec 12, 13)	Hold heights astride and beside the main axis; posts at platoon level; tiered defence (valley, mid slope, ridge); strong point on the most dominating feature; reverse slopes when the forward slope is under fire.	Posts: tier and strong point. Enemy Observation.
Observation (Sec 4, 22)	Dead space at mid ranges. OPs laterally, in depth and at varying heights so cloud cannot blind them all. Voices carry 900 m up: listening posts in valleys.	Posts: height tiers, dead ground. Viewshed.
Obstacles (Sec 12, 13)	Cover gaps with mines, fire and wire; mines thickest in valleys and foothills; mines wash away on slopes and fail in snow and rock: use wire there.	Minefields: valley bonus, steep and snow factors, notes.
Fire (Sec 18)	Up- or downhill fire strikes high: sight = slant range ÷ factor. Plunging fire leaves dead ground. MG mount +6° / −8.5°; 75 mm RR +30° / −5°. Mortars cover dead space.	Weapon sites: sight factor, mount limits. LOS: sight setting.
Artillery and AD (Sec 22)	High-angle weapons beat crests; light guns and mortars on reverse slopes and in valleys; harass passes and defiles. AD covers defiles, gun areas and logistic sites from heights.	Indirect fire: crest clearance, defilade. AD air corridors.
Advance (Sec 6)	The axis runs through valleys and defiles under flanking heights: picquet them for the main body and supply route.	Route Picquets.

Subject	GSP teaching	Where in MtnAI
Helicopters (Sec 22)	Landing sites ≤ 4° for regular use; emergency 4–10° light and 4–20° cargo. Hide site and approach. Hard ground, rock or grass.	Helipad Sites.
Animal transport (Annex H)	A loaded mule climbs any slope a man can without using his hands; about 16 miles a day, 8 under load; 65 kg in hills.	Mule class: SLOW-GO up to 30°.

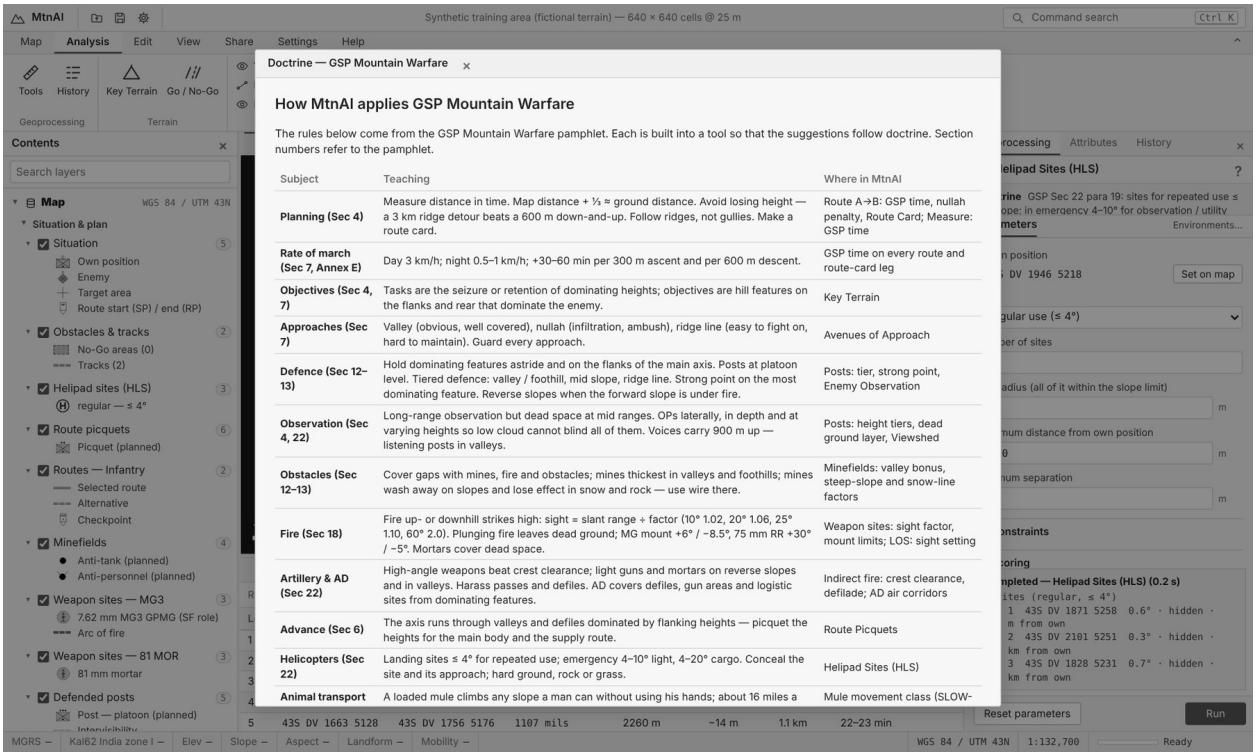


Figure 32. Help ▸ Doctrine inside the app.

14. SHORTCUTS AND LIMITS

Keys	Action
Ctrl+O	Add Data
Ctrl+S	Save plan
Ctrl+P	Report
Ctrl+,	Settings
Ctrl+K or /	Command search
Ctrl+G	Go To XY
Ctrl+B	Contents pane
Ctrl+T	Attribute table
Ctrl+J	Bottom dock
2 / 3	Map / Scene view
F	Full extent
+ / −	Zoom
M / L / V	Measure / Line of Sight / Viewshed
Esc	Cancel a tool, drawing or fly-through
F1	Help

Limits

- Results are only as good as the DEM. A 30 m DEM smooths cliffs, narrow nullahs and small features.
- Trees, snow depth, soil, rivers and buildings are not known unless you draw them.
- Weapon and movement figures are planning values. Use your unit's current data.
- Every suggestion is a starting point for reconnaissance, not a replacement for it.

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